

**SECURED COMMUNICATION MECHANISM
FOR MESSAGE AUTHENTICATION AND DATA
CONFIDENTIALITY IN SECS/GEM PROTOCOL**

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SECURED COMMUNICATION MECHANISM FOR MESSAGE AUTHENTICATION AND DATA CONFIDENTIALITY IN SECS/GEM PROTOCOL

by

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LIST OF ABBREVIATIONS

AE	Authenticated Encryption
AEAD	Authenticated Encryption with Associated Data
AES	Advanced Encryption Standard
AI	Artificial Intelligence
ASCII	American Standard Code for Information Interchange
CoAP	Constrained Application Protocol
CPS	Cyber-Physical Systems
DDS	Data Distribution Service
EEF	Engineering Employers' Federation
GCM	Galois/Counter Mode
GEM	Generic Equipment Model
GF	Galois Field
HSMS	High-Speed SECS Message Services
IBM	International Business Machines
IDE	Integrated Development Environment
IIoT	Industrial IoT
IoT	Internet of Things
IP	Intellectual Property
IT	Information Technology
Kbps	Kilobits per second
M2M	Machine-to-Machine
MQTT	Message Queuing Telemetry Transport
NIST	National Institute of Standards and Technology
OPC UA	Open Platform Communications Unified Architecture
OSI	Open Systems Interconnection
OT	Operational Technology
PLC	Programmable Logical Controller
RFC	Request For Comments
SECS	Semiconductor Equipment Communication Standard
SECS-I	SEMI Equipment Communications Standard, Part 1
SECS-II	SEMI Equipment Communications Standard, Part 2

SEMI	Semiconductor Equipment and Material International
SHA	Secure Hash Algorithm
SSH	Secure Shell
SSL	Secure Sockets Layer
TCP	Transmission Control Protocol
TCP/IP	Transmission Control Protocol / Internet Protocol
TLS	Transport Layer Security
TSMC	Taiwan Semiconductor Manufacturing Company
VoIP	Voice over IP
VPN	Virtual Private Network
WFH	Work From Home

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Appendix A	SECS/GEM-AE Host-S Processing Time
Appendix B	SECS/GEM-AE Equip-R Processing Time

MEKANISME KOMUNIKASI TERSELAMAT UNTUK PENGESAHAN MESEJ DAN KERAHSIAAN DATA DALAM PROTOKOL SECS/GEM

ABSTRAK

Industri 4.0 telah membawa revolusi dalam sektor pembuatan, yang telah menyebabkan peningkatan ketara dalam proses pembuatan dan peningkatan dalam kualiti dan kapasiti pengeluaran. Protokol komunikasi Mesin-ke-Mesin (M2M) telah dibangunkan untuk membolehkan mesin berkomunikasi dalam ekosistem perindustrian. Protokol komunikasi M2M iaitu Piawaian Komunikasi Perkakasan Semikonduktor/Model Peralatan Generik (SECS/GEM) telah wujud dalam industri pembuatan untuk tempoh yang agak lama, berfungsi sebagai protokol komunikasi dan sistem kawalan. Walaupun penggunaannya meluas, SECS/GEM tidak mempunyai ciri keselamatan seperti pengesahan mesej dan kerahsiaan, kerana ia direka untuk digunakan dalam rangkaian tertutup. Ketiadaan komponen keselamatan dalam protokol ini boleh membenarkan penggodam mencuri maklumat sulit, seperti proses pembuatan, dengan memeriksa parameter dan resipi peralatan. Mereka juga mungkin mengganggu atau mensabotaj komunikasi SECS/GEM, yang boleh memberi kesan buruk kepada industri yang menggunakan protokol tersebut dalam persekitaran mereka. Penyelidikan ini mencadangkan SECS/GEM-AE, iaitu suatu mekanisme untuk melindungi mesej data SECS/GEM dengan menggunakan penyulitan AES-GCM, dan menilai prestasi penggunaannya dalam protokol SECS/GEM asal serta SECS/GEMsec, mekanisme keselamatan yang telah dicadangkan untuk mengesahkan mesej SECS/GEM. SECS/GEM-AE berjaya memastikan kerahsiaan dan ketulenan data, dengan overhead yang amat rendah dan boleh diabaikan iaitu 0.45 milisaat dan 0.58 milisaat apabila menghantar dan menerima mesej, berbanding protokol asal.

Dalam perbandingan dengan SECS/GEMsec pula, SECS/GEM-AE menunjukkan pengurangan sebanyak 18.13 milisaat untuk menghantar mesej manakala 19.75 milisaat untuk menerima mesej. Kesimpulannya, SECS/GEM-AE adalah mekanisme yang dapat meningkatkan keselamatan untuk komunikasi SECS/GEM tanpa overhead yang tinggi.

**SECURED COMMUNICATION MECHANISM FOR MESSAGE
AUTHENTICATION AND DATA CONFIDENTIALITY IN SECS/GEM
PROTOCOL**

ABSTRACT

Industry 4.0 has brought about a revolution in the manufacturing sector, resulting in significant enhancements in the manufacturing process and an increase in production quality and capacity. Machine-to-Machine (M2M) communication protocols have been developed to bind the industrial ecosystem, enabling machines to communicate. The Semiconductor Equipment Communication Standard/Generic Equipment Model (SECS/GEM) M2M communication protocol has been at the forefront of the manufacturing industry for years, serving as both a communication protocol and control system. However, despite its widespread adoption, SECS/GEM lacks security features such as message authentication and confidentiality, as it was designed for use in closed networks. The deficiencies in the protocol can permit malevolent actors to steal confidential information, such as manufacturing processes, by examining the equipment parameters and recipes. These individuals may also disrupt or sabotage SECS/GEM communications, resulting in grave consequences for the industry. This study proposes SECS/GEM-AE, a mechanism for securing SECS/GEM data messages using AES-GCM encryption and evaluates its performance against the standard SECS/GEM protocol and SECS/GEMsec, a security mechanism proposed to authenticate SECS/GEM messages. SECS/GEM-AE successfully provides message authentication and confidentiality, with a negligible overhead of 0.45 milliseconds and 0.58 milliseconds when sending and receiving messages, respectively, compared to the standard protocol. When compared to SECS/GEMsec,

SECS/GEM-AE shows a reduction of 18.13 milliseconds in sending messages and 19.75 milliseconds in receiving messages. The results conclude that SECS/GEM-AE is the appropriate security mechanism to secure SECS/GEM communications.

CHAPTER 1

INTRODUCTION

This chapter is divided into nine sections. Section 1.1 presents an overview of Industry 4.0, its history, and the significance of Machine-to-Machine (M2M) communication protocols in modern industrial networks. Section 1.2 provides an overview of the SEMI Equipment Communications Standard/Generic Equipment Model (SECS/GEM) M2M communication protocol. Section 1.3 introduces industrial networks and the cybersecurity threats in modern industrial networks. Section 1.4 presents the motivation for this research. Sections 1.5 through 1.9 present the problem statement, research objectives, scope and limitations, research contribution, and research steps. Finally, Section 1.10 provides the organization of the rest of this thesis.

1.1 Overview

The First Industrial Revolution was the era in the 18th century when Europe, America, and Great Britain adopted novel industrial practices (Lu, 2017; Sanchez et al., 2020). This transition encompassed the transfer from manual handcrafting to machine production, the establishment of innovative iron and chemical production practices, the increased utilization of hydropower and steam engines, industrial machinery, and the factory system (Mourtzis et al., 2022). Nevertheless, industrialization in the 18th century resulted in an astonishing acceleration of production expansion. The advent of electrical technology in the mid-nineteenth century led to significant scientific breakthroughs and standardization, which echoed the objective of mass production; hence, it was termed the Second Industrial Revolution (Suleiman et al., 2022). The industries took off once the Programmable Logical Controller (PLC) was invented in late 1960, revolutionizing industrial

automation and coining the phrase Third Industrial Revolution (Alay, 2023). The rapid progression of technology in recent decades has led to the emergence of Industry 4.0, also known as the Fourth Industrial Revolution. This is due to the integration of various cutting-edge technologies such as machine learning, Artificial Intelligence (AI), robotics, Internet of Things (IoT), 5G networks, cloud computing, 3D printing, and quantum computing (Alhayani et al., 2023).

The manufacturing industry has been revolutionized by Industry 4.0, vastly improving the manufacturing process and increasing production quality and capacity. M2M communication protocols were developed to strengthen and bind this ecosystem by allowing machines to communicate with each other (Leang & Rasiah, 2023). With sophisticated machinery and automation, more integrated M2M communication, real-time monitoring and data collection, machine learning, and enhanced interconnectivity, Industry 4.0 is changing the existing manufacturing process for better and improved overall production (Zheng et al., 2021). Interconnected machines generate activity information, predictive diagnostic data, performance statistics, and other monitoring and control information. Thus, real-time decisions can be made quickly with advantages such as time and cost-saving. In many scenarios, human involvement in the factory environment can be eliminated. With the utilization of pre-set and tested settings and parameters, factory equipment can make critical decisions autonomously, thereby ensuring optimal cost-effectiveness for the industry.

M2M communication is a critical component of Industry 4.0, enabling the creation of smart factories and a more connected, automated, and efficient industrial landscape. M2M communication protocols allow machines to exchange data and control information in real-time, enabling them to operate independently and integrate

seamlessly with the factory environment. M2M communication enables machines to process and act on data, make informed decisions, and take autonomous actions to improve productivity and efficiency. M2M communication lies at the heart of Industry 4.0, which possesses the potential to revolutionize the production of goods and services by streamlining manufacturing processes and reducing the requirement for manual intervention. In addition, using M2M communication protocols can help companies gain a competitive advantage by enabling them to respond to changes in market demands and customer needs with greater agility and flexibility.

Industry 4.0 involves a lot of M2M communications and processing of sensitive data, and security is a critical concern that must be addressed to fully realize its potential. The increased data density associated with Industry 4.0, along with the convergence of information technology (IT) and operational technology (OT), brings new challenges, particularly in the area of cybersecurity (Ervural & Ervural, 2018). Cybersecurity is a top priority for governments worldwide, as it involves protecting digital business information and valuable subject or system knowledge against abuse, unauthorized access, and theft (Buch et al., 2017). Figure 1.1 presents the percentage of extortion cases per industry, due to cybersecurity incidents in 2022, observed by IBM X-Force (Worley et al., 2023) . It is reported that attacks on the manufacturing sector have significantly increased which has made manufacturing industry to lead the list.

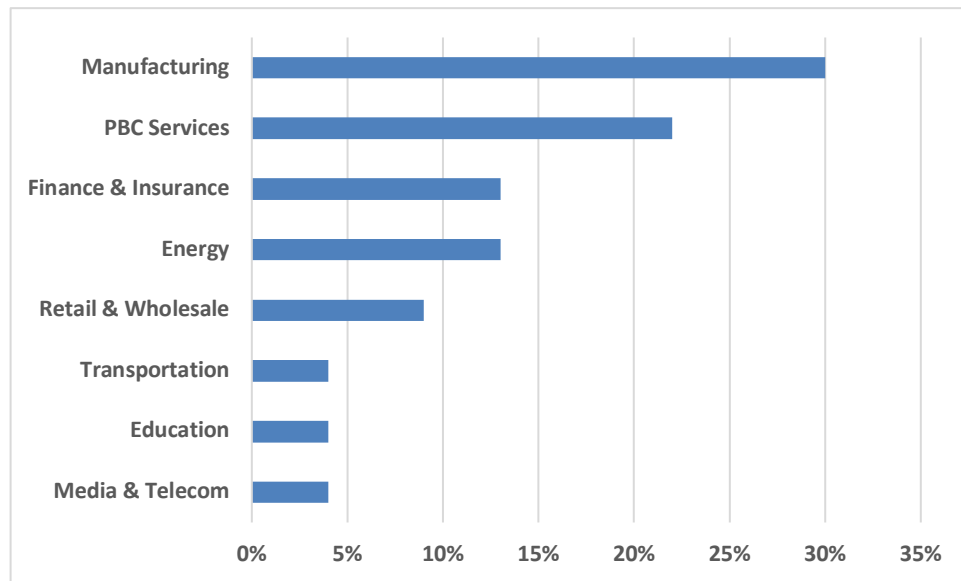


Figure 1.1 Extortion Cases Observed by IBM X-Force in 2022 (IBM, 2022)

With expanding network connections, cyberattacks have become more prevalent due to the rising tendency to misuse data for different purposes, such as financial and strategic reasons (Ervural & Ervural, 2018). Therefore it is essential to implement robust security features to avoid various cyber-attacks (Kwon et al., 2020).

The industry utilizes various M2M communication protocols, including SECS/GEM, Data Distribution Service (DDS), Open Platform Communications Unified Architecture (OPC UA), and Message Queuing Telemetry Transport (MQTT), which are among the most commonly employed protocols. Figure 1.2 shows popular IoT and Industrial IoT (IIoT) Protocols.

In the realm of the Internet of Things (IoT) and Industrial Internet of Things (IIoT), various protocols play pivotal roles in facilitating communication and data exchange between connected devices. In the domain of IoT, protocols such as XAMP, LwM2M (Lightweight M2M), and CoAP (Constrained Application Protocol) are commonly employed. XAMP provides a robust framework for real-time data streaming, while LwM2M is designed for efficient device management and CoAP, a

lightweight protocol, is optimal for resource-constrained devices. In the IIoT landscape, protocols like OPC UA (Unified Architecture), AMQP (Advanced Message Queuing Protocol), DDS (Data Distribution Service), and SECS/GEM (SEMI Equipment Communications Standard/Generic Equipment Model) take precedence. OPC UA ensures interoperability in industrial automation, AMQP facilitates efficient message queuing, DDS supports real-time data distribution, and SECS/GEM is vital in semiconductor manufacturing. Bridging the domains of IoT and IIoT, protocols such as Zigbee and MQTT find application. Zigbee is prominent in low-power, short-range communication, while MQTT is widely adopted for lightweight and efficient messaging in both industrial and general IoT scenarios. These protocols collectively contribute to the seamless integration and functionality of connected devices across diverse applications in the IoT and IIoT ecosystems.

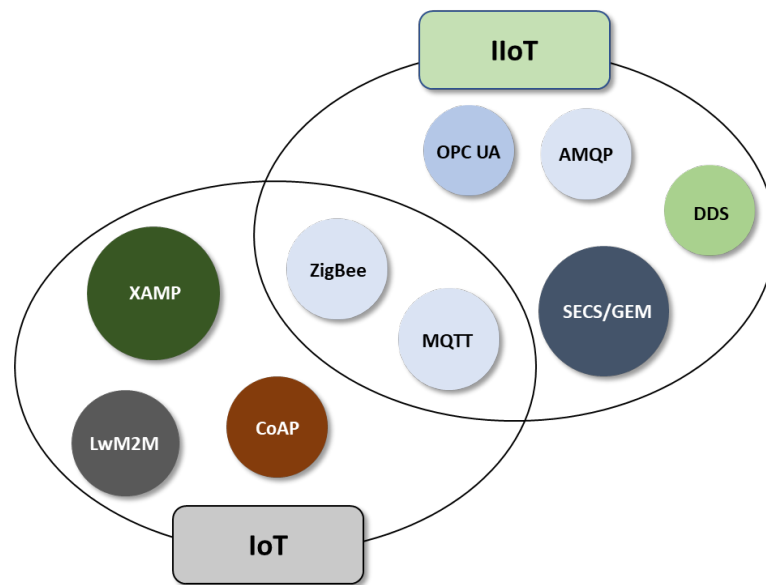


Figure 1.2 IoT and IIoT Communication Protocols

1.2 SECS/GEM Protocol – An overview

SEMI (formerly Semiconductor Equipment and Materials International) has introduced numerous standards, including four major protocol standards for

equipment-to-host communications (Ehm et al., 2020). These standards were first released in 1978, with the latest revision published in 2020. Though the SECS/GEM communication protocols were published several years ago, they are regularly maintained and updated. This section is an overview of the major SECS/GEM protocol releases. Table 1.1 gives a brief description of SECS/GEM standards.

Table 1.1 SECS/GEM Standards

Year	Standard	Description
1978	E4 SECS-I	SEMI Equipment Communications Standard-I is a communication protocol for establishing communication between equipment and a host via RS-232 cable at the physical layer of the network stack.
1982	E5 SECS-II	SEMI Equipment Communications Standard-II enables the exchange of information between equipment and hosts using streams and function messages in a defined format.
1992	E30 GEM	The Generic Equipment Model defines SECS-II message usage and monitors equipment behavior during message exchange with the host.
1994	E37 HSMS	High-Speed SECS Message Service is a protocol for managing point-to-point communication between equipment and hosts over TCP/IP.

The SECS/GEM protocol is a point-to-point M2M communication protocol, meaning factory equipment is connected to a single host machine and only communicates with the mentioned host machine. Host systems are connected to various equipment that must be SECS compliant (i.e., based on SECS standards). Serial (RS-232) or TCP/IP ports can be used to connect to the host. The GEM standard is used in some of the more recent equipment. Generally, older equipment is not GEM compliant, relying on serial connectivity. The HSMS standard operates

similarly to TCP but does not support multiple connections. A piece of equipment can only connect to and communicate with one host at a time and must disconnect from the current host before communicating with another. Figure 1.3 illustrates an implementation architecture for a SECS/GEM-enabled factory system.

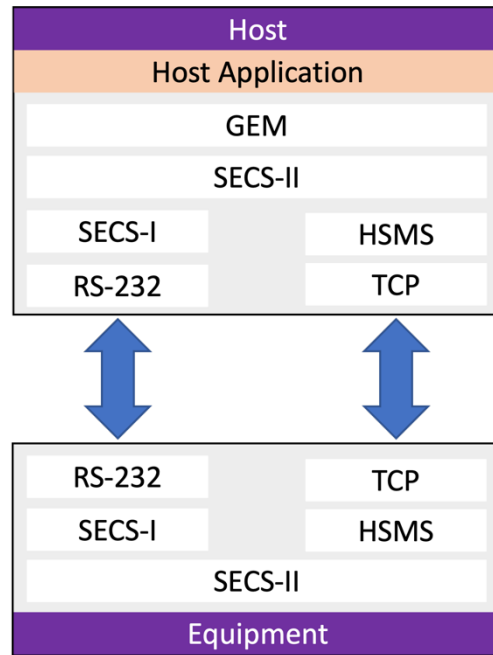


Figure 1.3 Implementation Architecture for SECS/GEM (Azaiez et al., 2019)

The host system's primary function is to manage connectivity to a group of equipment via a configuration file, recipe management, load balancing, monitoring of the equipment, and connection to enterprise-level applications. SECS/GEM messages are sent as requests and responses. For example, a host may request a piece of equipment for data, and the equipment will reply with the requested data as a response. A request message and a response message together are called a transaction. Every message has a detailed header that states the message type and its function. The SECS/GEM message structure is covered in more detail in Chapter 2.

Although SECS/GEM has been widely adopted (Ewe et al., 2020; Zhu et al., 2021), the standard does not offer any security mechanism to secure communications.

The SECS/GEM communications are transmitted as binary encoded data. An attacker with fair knowledge of SECS/GEM and binary encoding can easily read the contents of these messages. This can lead to serious problems such as intellectual property (IP) theft. Moreover, SECS/GEM does not authenticate or verify the messages' integrity and will accept and process illegitimate or modified messages. This vulnerability allows attackers to manipulate equipment with very minimal effort.

Despite the widespread use of the SECS/GEM protocol in industrial networks, little research has been conducted on securing these communications. Specifically, no work has been published to address confidentiality issues in SECS/GEM, which raises significant concerns about the security of this widely used protocol. This lack of attention has prompted researchers to develop alternative methods for securing SECS/GEM communications, such as the digital signature-based approach known as SECS/GEMsec (S. U. A. Laghari, Manickam, Al-Ani, et al., 2021a). While SECS/GEMsec offers message authentication, it fails to address other important aspects, such as data confidentiality and performance. SECS/GEMsec is covered in detail in Chapter 2.

1.3 Industrial Networks and Cybersecurity Threats

Industry 4.0 emphasizes the use of automation and remotely controllable operations. In the past, manufacturing environments were not connected to external networks, so ensuring cybersecurity was not a major concern. However, protocols were not secured even when the equipment was connected, as manufacturing networks were typically closed and air-gapped environments where external actors posed little to no threat. The drive of automation and Cyber-Physical Systems (CPS) in the manufacturing environment requires the manufacturing network to be open, therefore

requiring the evaluation of those protocols being used and addressing any forthcoming issues, especially from a security perspective.

Cyberattacks pose threats to the basic cybersecurity principles, i.e., confidentiality, integrity and availability (Brar & Kumar Ahuja, 2018). When evaluating protocols for use in Industry 4.0 environment, these cybersecurity principles are required to be followed. Figure 1.4 depicts a taxonomy of cyberattacks on cybersecurity principles (Brar & Kumar Ahuja, 2018). Attacks on cybersecurity principles (confidentiality, integrity, and availability) pose a major threat to cybersecurity. Therefore, these principles must be ensured for a secure Industry 4.0 enabled manufacturing environment.

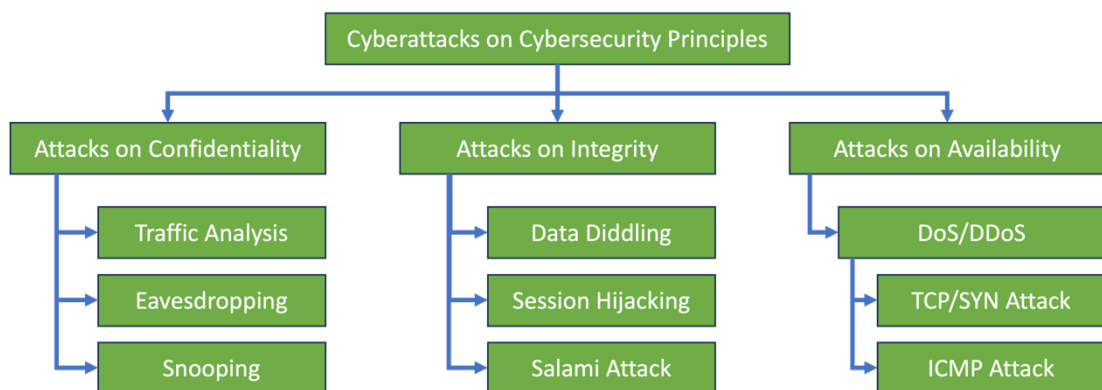


Figure 1.4 Taxonomy of Cyberattacks on Cybersecurity Principles

The manufacturing industry has been gradually updating and improving its IT security over the years; however, it can be seen in the Verizon Data Breach Investigation Report 2019 (Widup, 2019), detailing 352 cyberattack incidents, out of which 87 were against the manufacturing industry. Recent attacks and security breaches against the manufacturing industry are alarming, making it a highly targeted and vulnerable entity for attackers (Tuptuk & Hailes, 2018). A survey by the Engineering Employers' Federation (EEF) shows that 60% of manufacturers were victims of cyberattacks at some point, and one-third of the affected manufacturers have

suffered financial and market losses due to these cyberattacks. According to a study conducted by Cybersecurity Ventures in 2021, corporations across the globe are expected to experience annual losses of up to \$10.5 trillion by 2025 due to cyberattacks. This projection is a significant increase from the estimated losses of \$3 trillion in 2015 (Morgan, 2021). The cyberattack on Taiwan Semiconductor Manufacturing Company (TSMC) was, in Taiwan's history, the worst data security infringement to befall them. It completely exposed data security vulnerabilities at TSMC's production foundries. These cyber-attack incidents are happening as the manufacturing industry embraces the shift to Industry 4.0, with more and more machines becoming connected for communications and automation (Peng, 2018). Figure 1.5 depicts the distribution of cyber-attacks across leading industries worldwide in 2022 (Worley et al., 2023). It can be seen that the manufacturing industry suffered the greatest number of cyber-attacks and is expected to grow even further in the future.

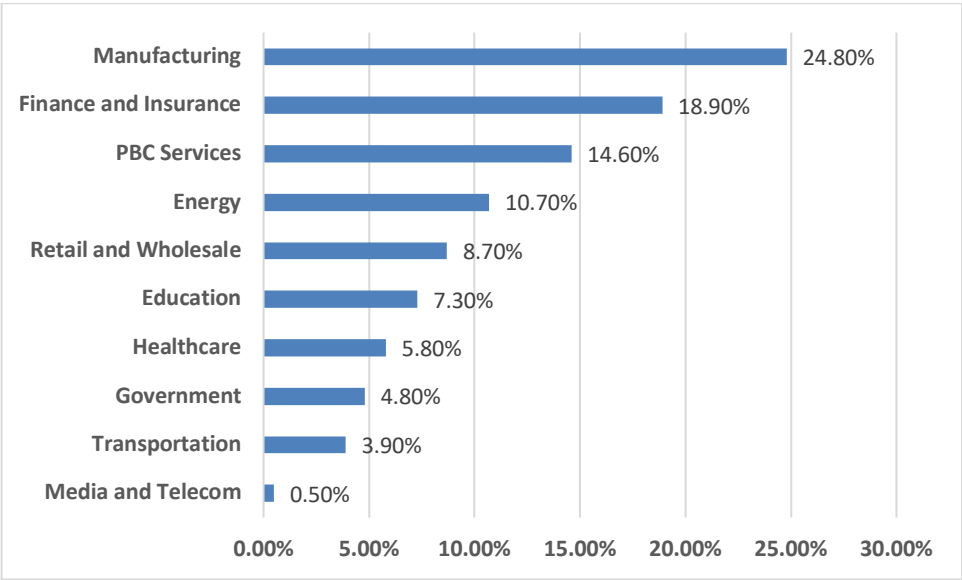


Figure 1.5 Cyber-Attacks on Industrial Sectors in 2022 (Worley et al., 2023)

At the end of 2019, the Corona Virus Disease 2019 (COVID-19) pandemic took the world by storm (Georgiadou et al., 2022). The process to stop the spread of the pandemic came with its consequences, requiring governments to impose

lockdowns and suspend operations in sectors that were deemed non-essential. The manufacturing industry came under these restrictions as well. The stakeholders had no choice but to allow their employees to work remotely and only have minimal staff to ensure their safety. This led to the Work From Home (WFH) scenario allowing employees to connect to corporate and manufacturing networks and work from home (Georgiadou et al., 2022). This scenario opened the door for cybercriminals to infiltrate manufacturing industries even from outside manufacturing facilities, rendering them vulnerable to even more cyber threats.

The cybercriminals knew of the WFH situation and only needed to break into an employee's computer to access manufacturing networks (Khatri et al., 2023). Thus in 2020, when many industries relied on their remote workers, cybercriminals took advantage of the situation to intensify their attacks. The manufacturing industry jumped from the eighth to the second most targeted industry by cybercriminals, with a 300 percent increase in cyber-attacks in a year (Kazu & Thomas, 2021).

1.4 Research Motivation

Industrial networks are increasingly vulnerable to attacks that target business secrets and intellectual property. This issue is particularly pressing in the semiconductor industry, where the SECS/GEM protocol is widely used for communication between equipment and manufacturing control systems. However, the plaintext nature of this protocol renders it susceptible to passive attacks. Since existing security mechanisms are not suitable for addressing these vulnerabilities, and if they are available, they tend to be complex and require excessive processing time for messages. Consequently, there is a crucial need for a security mechanism that can protect business secrets and maintain confidentiality in real-time. This mechanism

must also operate at high efficiency to avoid slowing down the system and impeding productivity.

Given the rapid growth of the semiconductor industry, it is essential to develop a robust and reliable security mechanism that can prevent attacks and safeguard sensitive information. By addressing the security flaws of the SECS/GEM protocol, businesses can continue to operate with the necessary confidence and security. Therefore, this research endeavors to propose a security mechanism that is not only fast and efficient but also provides real-time insights into the system, ensuring the confidentiality and protection of business secrets.

1.5 Problem Statement

Industry 4.0 has revolutionized the manufacturing industry and improved the manufacturing process, increasing quality and yield (Mourtzis et al., 2022). The same applies to the semiconductor industry, with the usage of automated machinery, robotics, sensors, and computers improving the overall production rate while, at the same time, being less prone to flaws and defects as the human element is removed and replaced with high precision machinery (Coronado et al., 2022). The whole manufacturing line of machinery, robotics, and sensors are interconnected by computers controlling everything, adjusting machine parameters accordingly, providing necessary recipes when needed, and so on (Soori et al., 2023). The operations are carried out through M2M communications, typically between a centrally managed server and the equipment and sensors. Such communication is carried out through special communication protocols. SECS/GEM is a popular M2M communication protocol widely adopted in the semiconductor industry (S. A. Laghari, Manickam, Karuppayah, et al., 2021). Although SECS/GEM has been widely adopted,

the standard does not offer any security mechanism to secure communications (Al-Shareeda et al., 2022). Researchers have proposed security mechanisms for SECS/GEM communications; however, they prove inadequate to be used in a production environment as they do not address necessary security vulnerabilities in SECS/GEM.

The SECS/GEM protocol and existing works on security features for SECS/GEM suffer from the following key issues:

The SECS/GEM standards and existing works exchange data in plaintext which makes it vulnerable to intellectual property theft. The existing works incur that offer security for SECS/GEM protocol are compute-intensive and require significant processing time, thus rendering their applicability unsuitable for the real-world applications.

1.6 Research Objectives

The objectives of this research are to achieve the following:

1. To propose a security mechanism that provides message authentication and data integrity.
2. To propose a security mechanism that provides data confidentiality.
3. To propose a security mechanism with minimal overhead and high efficiency to provide performance close to the standard SECS/GEM.

1.7 Scope and Limitation

The scope of this thesis is limited to proposing a security mechanism for authenticating industrial devices communicating over SECS/GEM protocol and

providing data confidentiality to the information exchanged between the two devices over the industrial communication network. Table 1.2 summarizes the overall research scope for the proposed security mechanism.

Table 1.2 Research Scope

Item	Scope of Research
Environment	TCP/IP Network
Security Mode	Authentication, Confidentiality
OSI Target Layer	Application Layer, Transport Layer
Evaluation Metrics	Processing Time, Protocol Control Overhead

This research employs symmetric key encryption algorithms. The secret keys employed in the encryption process are stored locally on the communication entities. The distribution and secure storage of secret keys are beyond the scope of this study.

1.8 Research Contribution

SECS/GEM is one of the most powerful and widely implemented protocols in the manufacturing industry, yet its security features have only recently been realized. The current research focuses only on authentication and prevention of cyberattacks to enable using this protocol in the Industry 4.0 ecosystem. Hence, this study focuses on the authentication and confidentiality of SECS/GEM with a focus on performance. The forthcoming study is expected to make the following contributions:

1. The proposed mechanism enables SECS/GEM devices to encrypt communications, attain data confidentiality and prevent Intellectual Property (IP) theft.

2. The proposed mechanism secures SECS/GEM communications while maintaining performance acceptable in a production environment.
3. A testbed environment is developed to evaluate the performance of the proposed mechanism in terms of processing time and control overhead.

1.9 Research Steps

It is convenient to split the research into five stages or steps to attain this study's objectives. These phases are as follows: problem identification, literature review, research methodology, implementation, and evaluation. Figure 1.6 illustrates each of these stages/steps and associated activities to complete this study.

Step 1: Problem Identification. This stage covers the SECS/GEM communication protocol in general and delves deeper into the underlying SEMI communication standards to highlight the SECS/GEM protocol's strengths and weaknesses. Security threats to SECS/GEM communications caused by a lack of confidentiality support are also identified at this stage. The problem statement is defined based on the issues identified. The scope and limitations are established in this stage as well.

Step 2: Literature Review. This stage reviews existing security solutions available for SECS/GEM. Hence, this stage provides deeper insight into the limitations of existing solutions and the requisite knowledge to describe the proposed solution.

Step 3: Research Methodology. This stage describes the proposed mechanism with authentication and confidentiality features for SECS/GEM communications. The proposed mechanism must not change the existing SECS/GEM message structure. It

must also be efficient for encrypting and authenticating messages while maintaining acceptable performance.

Step 4: Implementation. This stage explains the specifics of how the proposed mechanism is implemented, the programming language, and the steps taken to execute the proposed mechanism in the testbed environment. Additionally, this phase discusses software tools for monitoring and capturing network traffic. The tools discussed will be used in conjunction with the development of the proposed mechanism. Additionally, a testbed will be developed to examine the functioning and efficiency of the proposed mechanism. The proposed mechanism is executed, and its performance is evaluated in various configuration modes.

Step 5: Evaluation. This stage is the analysis of the experiment results. The results are evaluated based on certain evaluation matrices and compared with existing works. The proposed mechanism is also evaluated to verify if all the objectives were attained. Future works and improvements for the proposed mechanism are also identified.

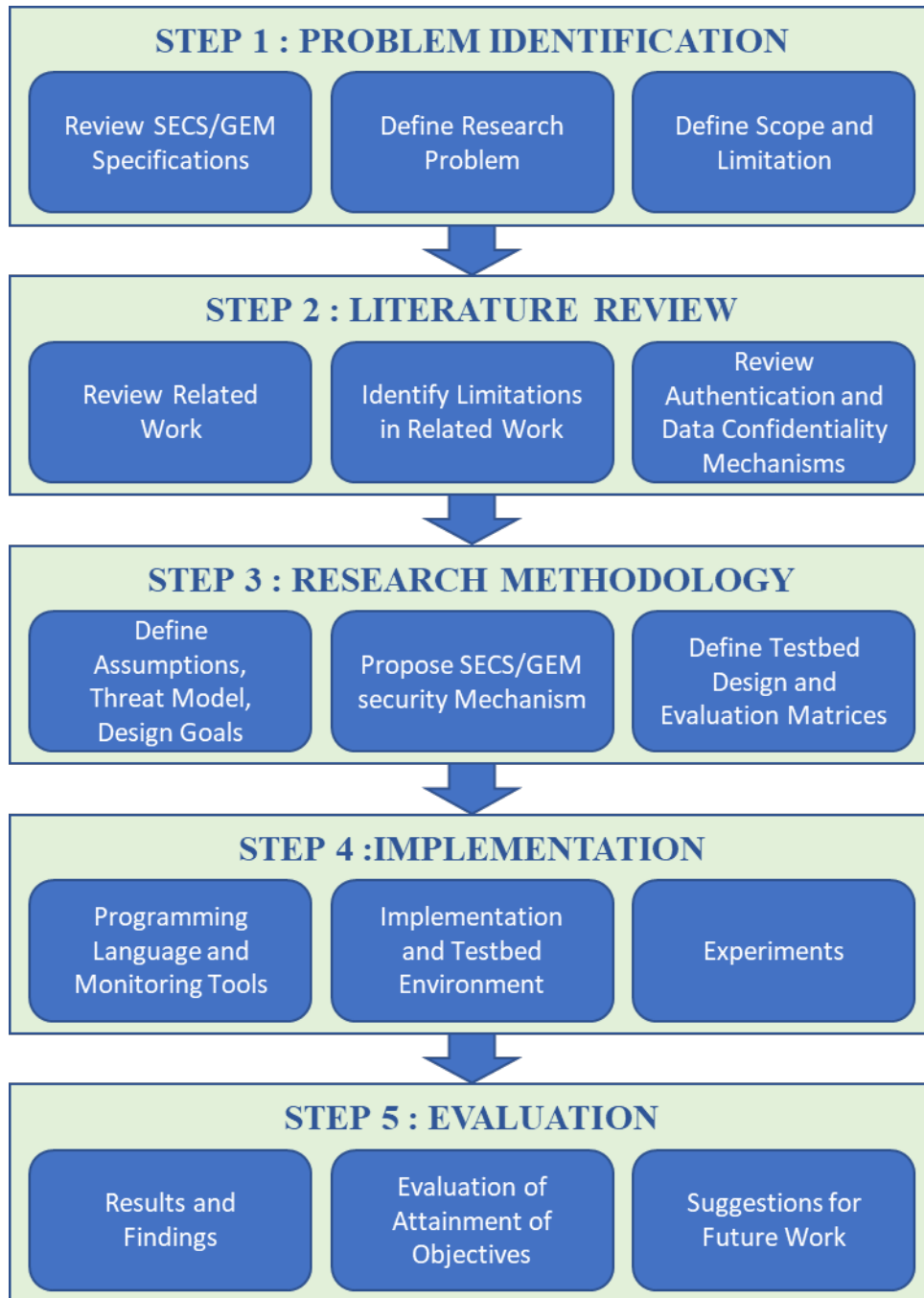


Figure 1.6 Research Steps

1.10 Thesis Organization

This thesis is organized into six chapters.

Chapter 1 introduces Industry 4.0 and the semiconductor manufacturing industry. An overview of M2M communication protocols and SECS/GEM M2M

communication protocol is also presented. The problem statement, research objectives, and research contributions are also established in this chapter.

Chapter 2 provides a background on SECS/GEM standards and cybersecurity issues associated with its protocols. It also reviews existing security mechanisms for SECS/GEM, their features, and shortcomings. It also reviews existing techniques that may be used to secure SECS/GEM communications.

Chapter 3 presents the methodology of this research work, detailing the features and requirements. It also proposes the design of a testbed to experiment with the proposed mechanism.

Chapter 4 presents the implementation of this research work, the programming language used, and the monitoring tools used. It also explains, in detail, the experiments, including variations and scenarios used in the experiments.

Chapter 5 presents the results, compares the results with existing works based on certain evaluation matrices, and discusses the findings of this research.

Chapter 6 summarizes this research work, provides suggestions for future work, and concludes this thesis.

CHAPTER 2

LITERATURE REVIEW

In this chapter, Section 2.1 presents a brief discussion of the SEMI organization and the manufacturing industry. Section 2.2 provides a comprehensive background of SEMI's SECS/GEM standards, namely SECS-I, SECS-II, HSMS, and GEM (also known as E4, E5, E37, and E30, respectively). The security issues of SECS/GEM are discussed in Section 2.3. Section 2.4 covers encryption and various types of encryption algorithms. In addition, this chapter reviews existing SECS/GEM communications security solutions and highlights their limitations in Section 2.5. The chapter is summarized in Section 2.6.

2.1 Background

SEMI is a global conglomerate of companies that focus on the semiconductor industry, comprising members worldwide. The organization consists of 26 subgroups: Manufacturing, Packaging & Test, Automated Test Equipment, Market Statistics, Information & Control, Health & Safety, and more (Goh et al., 2017). SEMI provides equipment, software, materials, and services to produce electronic devices, such as semiconductors, flat panel displays, photovoltaic cells, and other products (SEMI, 2020). Their market research provides valuable information on trends and advancements in semiconductors and related industries, encompassing market size, growth rates, and technological innovations. SEMI also aims to establish industry standards in the manufacturing, testing, and electronics supply chain to create universal guidelines. Additionally, SEMI supports policies and regulations that promote the growth and competitiveness of the semiconductor industry and others.

The utilization of automation in the semiconductor manufacturing industry has been gaining importance in recent times. Technologies such as robots, AI, and computer vision are being deployed to boost efficiency, cut costs, and improve the quality of products. According to recent studies, automation technologies in semiconductor industry can lower manufacturing costs while decreasing production time (Mane, 2022). Additionally, automation is helping to minimize environmental pollution, as automated systems can more accurately supervise and regulate semiconductor production processes. Furthermore, automation technologies could decrease waste and improve product quality (Javaid et al., 2021).

Figure 2.1 depicts simplified equipment configurations within an industrial production line network.

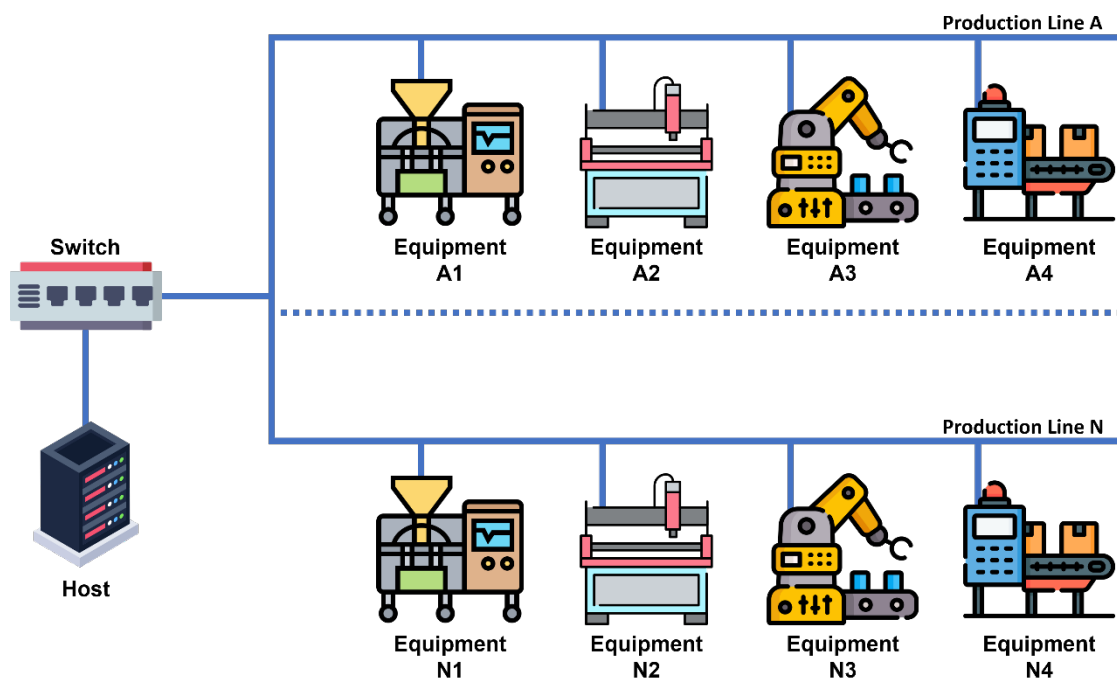


Figure 2.1 Simplified Industrial Production Network Topology

M2M communication has become a critical component in industrial networks, where numerous devices, sensors, and machines need to interact and share information to enable automation and optimize production processes. M2M communication allows machines to communicate with each other directly without human intervention,

reducing the need for manual monitoring and control. This type of communication can occur over various protocols. By leveraging M2M communication, industrial networks can achieve greater efficiency, accuracy, and speed, reducing costs and increasing productivity. Additionally, M2M communication can enable predictive maintenance and real-time monitoring of industrial assets, improving reliability and reducing downtime. In numerous cases, there will be no involvement of human intermediaries. Instead, machines can independently make critical decisions with predetermined and controlled parameters, thus optimizing cost-effectiveness for the manufacturer. The SECS/GEM communication protocol is discussed in this chapter based on relevant scientific research.

2.2 SEMI M2M Communication Standards

Semiconductor Equipment and Material International (SEMI) is a global association with over two thousand members that provides equipment, materials, and services to the manufacturing industry. It has created several standards, including E4, E5, E30, and E37, to facilitate communication between factory equipment and the host. These standards are collectively known as SECS/GEM, as shown in Table 1.1. The hierarchal representation of SECS/GEM standards are shown in Figure 2.2. SECS/GEM is an industry-standard used in various manufacturing industries for decades. In companies such as Intel, Samsung, TSMC, IBM, Qualcomm, Broadcom, UMC, SK Hynix, Micron, TXN, Toshiba, NXP, and others, SECS/GEM serves as a communication protocol and control system, making it an essential part of the semiconductor industry for many years.

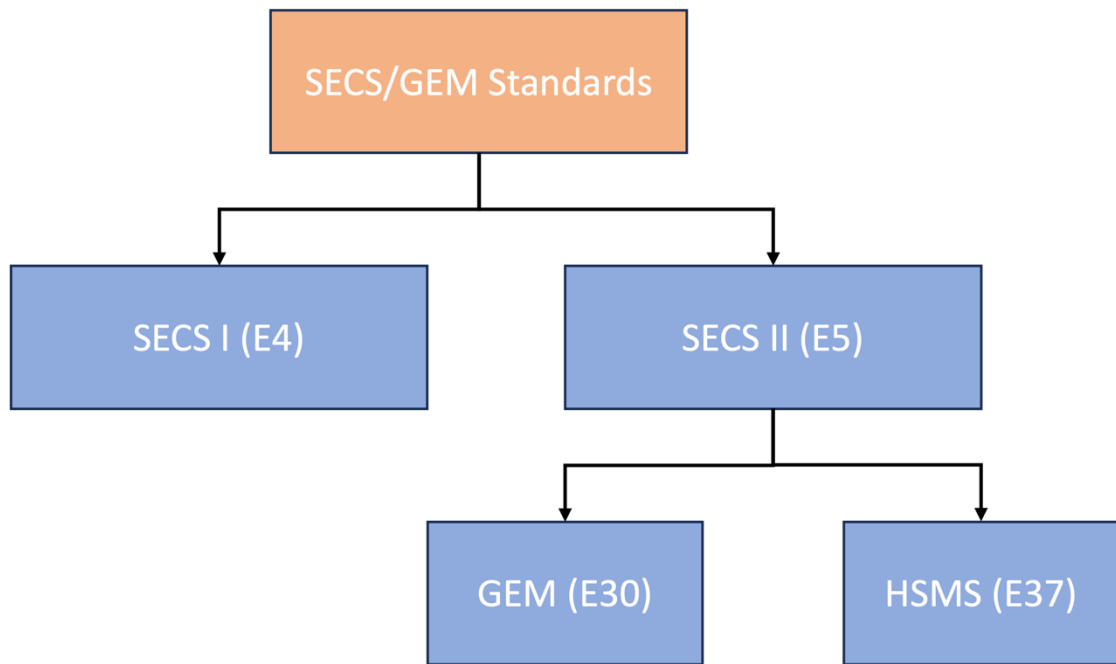


Figure 2.2 SECS/GEM Standards Hierarchal Representation

The SECS/GEM standard establishes a set of requirements for communication between semiconductor manufacturing equipment and host computer systems. SECS/GEM protocol defines a set of features referred to as *Fundamental Requirements* that equipment suppliers must implement in their devices to ensure compatibility with other equipment and host systems. Fundamental requirements are the basic features that are mandatory for all SECS/GEM-compliant devices, including status report messages, data collection, and event reporting. Following is the detailed list of fundamental requirements of SECS/GEM standard:

- Message structure: The SECS/GEM standard defines a specific message format that should be used for communication between equipment and host systems. This message structure includes fields for message type, equipment ID, transaction ID, and data.
- Message transfer protocol: The standard defines a protocol for transferring messages between equipment and host systems. This protocol includes

procedures for establishing and terminating communication sessions, sending and receiving messages, and error handling.

- State model: The SECS/GEM standard defines a state model for equipment that specifies the various states that equipment can be in during manufacturing. This state model provides a common language for equipment and host systems to communicate about the status of equipment.
- Alarms and events: The standard defines a set of alarm and event messages that can be used to notify host systems of equipment status changes or errors. These messages can trigger automated responses or alert operators to take corrective action.
- Data collection: The SECS/GEM standard defines a set of data collection messages that can be used to request specific information from equipment or to send data to equipment for processing.

The SECS/GEM standard provides a standardized framework for communication between semiconductor manufacturing equipment and host computer systems, which helps to ensure interoperability and compatibility between different systems.

In addition to the fundamental GEM requirements, SECS/GEM interface can support several optional capabilities depending on the specific implementation and requirements of the equipment and host systems involved. Some of the optional capabilities that a GEM interface can support include the following:

- Remote Command and Control: This allows the host system to send instructions to the equipment and carry out operations remotely.
- Recipe management: This capability enables the host system to manage the equipment's recipes for every manufacturing process.

- **Traceability:** This capability enables the equipment to provide data about the manufacturing process that can be used to trace products through the production processes.
- **Data monitoring:** This capability enables the host system to collect and monitor data from the equipment to analyze performance, identify trends, and make informed decisions.
- **Dynamic allocation of resources:** This capability enables the equipment to adjust its resource allocation based on real-time data and conditions to improve efficiency and optimize performance.
- **Automatic material handling:** This capability enables the equipment to handle and process materials used in manufacturing automatically, thus, reducing the need for manual intervention and increasing productivity.

The optional capabilities supported by a GEM interface depend on the specific needs and requirements of the equipment and host systems involved and can vary widely depending on the implementation.

2.2.1 SEMI Equipment Communications Standard 1 (SECS-I)

SEMI E4 standard, or SECS-I, is a communication protocol used in the semiconductor industry to enable communication between equipment and factory control systems. The SECS-I standard defines the physical and electrical interface for data transfer between devices, including message transmission, data structures, and message sequences. This standard is based on RS-232 serial communication protocol and operates at a speed of up to 57.6 kilobits per second (kbps). SEMI developed SECS-I to standardize communication protocols in the industry, improve equipment efficiency, and reduce downtime. Despite its speed and data capacity limitations, SECS-I remains an important standard in the semiconductor industry, particularly for older

equipment that may not support newer communication protocols. However, the SECS-I protocol is not bundled with new machines. Thus, this research disregards SECS-I in favor of HSMS, a TCP/IP-based transport protocol for SECS/GEM.

2.2.2 SEMI Equipment Communications Standard 2 (SECS-II)

SECS-II, also known as the SEMI E5 standard, is a communication protocol that utilizes a message-based format. It defines a general message layer that can transmit any data structure specified by the standard and a set of standard messages with unique identities, purposes, and formats. SECS-II provides a framework for interpreting message types, data types, message structure, and contents exchanged between factory equipment and a host. The messages are grouped into categories called streams, each of which covers a specific group of procedures. Streams are further divided into functions, which are specific messages within a particular stream category. A combination of a stream number and a function number is used to identify each message, for example, S1F13, where S1 represents stream one and F13 represents function thirteen. Primary messages (request messages) have odd-numbered function codes, while secondary messages (reply messages) have even-numbered function codes. Transactions consisting of a primary and secondary message are identified by a unique transaction ID called SystemBytes and initiated by either the host or the equipment. The sender sets a SystemBytes value to pair the primary message with its corresponding secondary message. Figure 2.3 illustrates a connection establishment transaction.

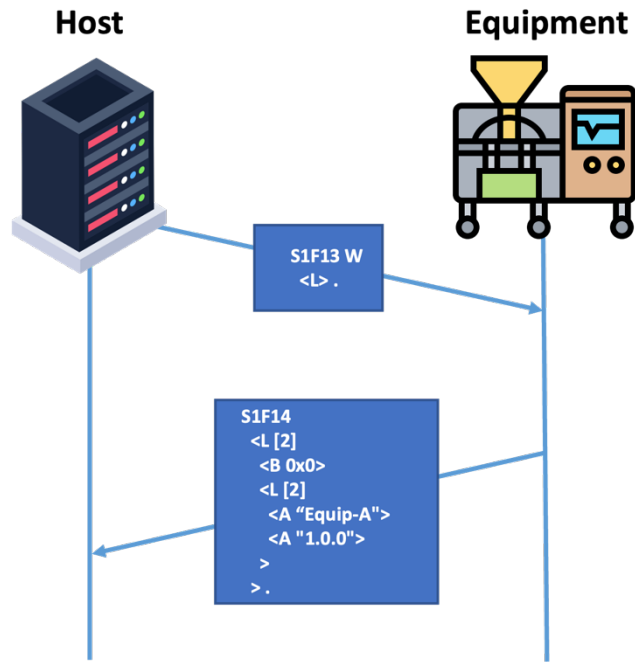


Figure 2.3 Connection Establishment Transaction

Figure 2.4 depicts a simple connection establishment request message, S1F13, from the host. The message contains an empty list with the W-bit set, indicating that the host expects a reply message from the equipment. The message size is calculated based on the data passed from the SECS-II layer. Since the data portion consists of an empty list, the payload size is only 2 bytes. The total message size is 16 bytes, which includes length (4 bytes), header (10 bytes), and data (2 bytes). The list data item in the message has a format code and length field necessary for encoding data.

In response to the S1F13 message, the equipment will send a reply message, S1F14, that includes the required information in the data portion. Figure 2.5 illustrates the reply message, which contains two lists, one binary data item, and two ASCII data items. The binary encoded packaging of the data portion is also depicted. It should be noted that the presence of a list data item increases the payload size, which can negatively impact the data density.

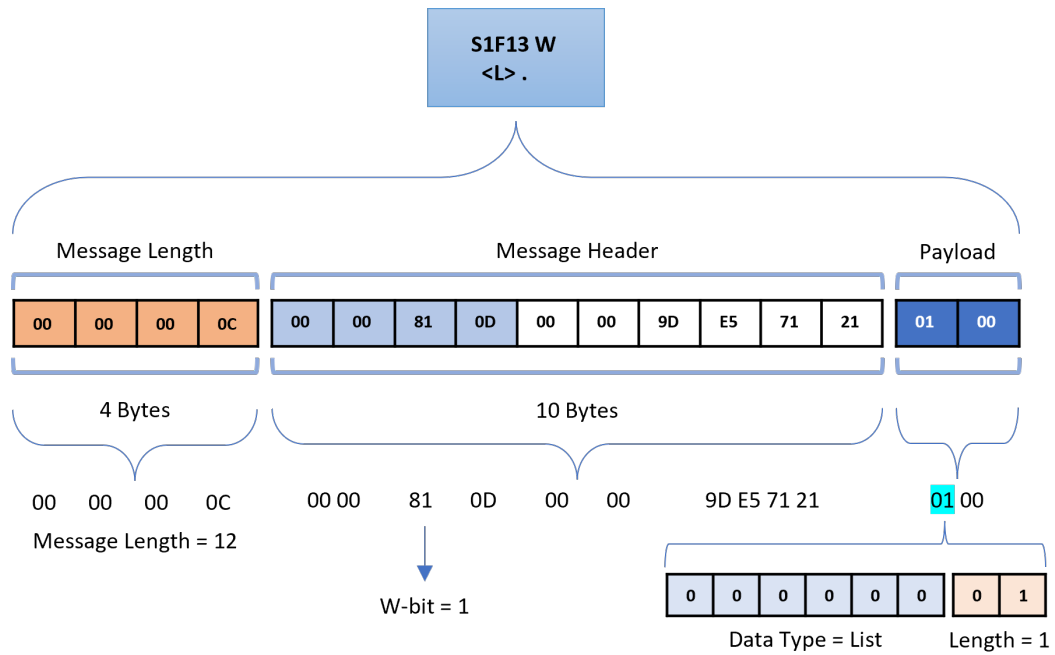


Figure 2.4 Connection Establishment Request Message Structure

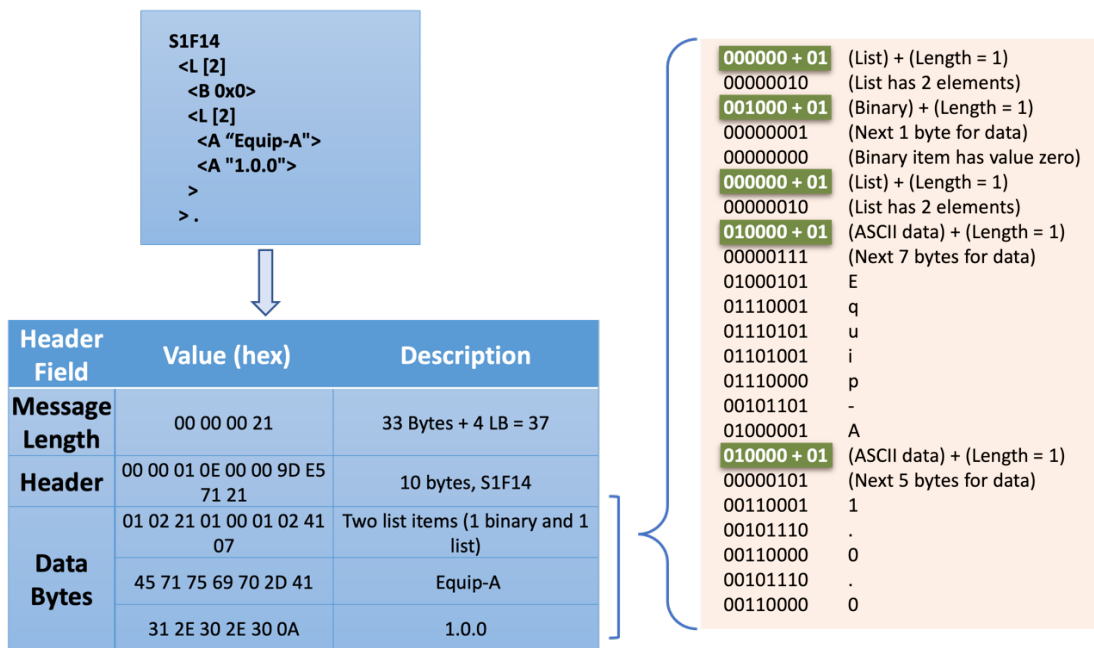


Figure 2.5 Connection Establishment Response Message Data Packaging

SECS-II reply message S1F14 is presented and displayed in Wireshark in Figure 2.6. The image shows that the message includes a list of two items. Interestingly, while the HSMS protocol indicates a packet length of 33 bytes, TCP shows a 37-byte payload. The HSMS packets are all prefixed with 4 bytes, which determine the packet length, and these bytes are stripped off during packet processing at the HSMS level.

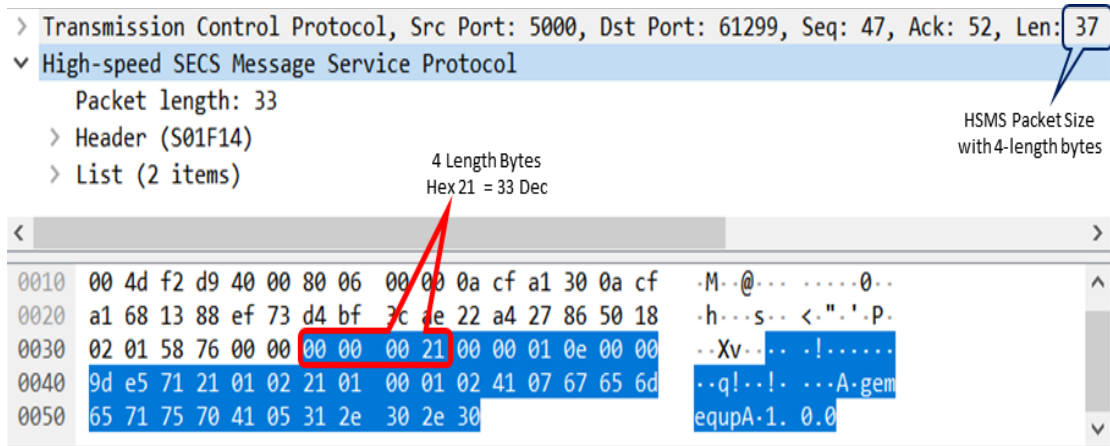


Figure 2.6 Wireshark Capture S1F14

SECS-II utilizes simple data types to encode messages in a very efficient and compact manner. These data types, descriptions, and sizes are outlined in Table 2.1. The maximum size of a single data element within SECS-II messages is restricted to 16,777,215 bytes, equivalent to about 16.5MB. A message may contain a basic data element such as a binary response or an ASCII string, a complicated list structure that includes multiple levels of lists in a hierarchy, or no data.

Table 2.1 SECS-II Data Types

Data Type	Size (bytes)	Description
Signed Integer	1, 2, 4, 8	Used to represent signed numbers (positive and negative)
Unsigned Integer	1, 2, 4, 8	It is used to represent whole numbers
Boolean	1	It is used to represent on/off values
ASCII	~	It is used to describe strings
Binary	~	It is used to store binary data such as images and graphs
List	~	It represents a list of data items. It can contain a sequence of other primitive data items and nested lists.

2.2.3 High-Speed SECS Message Services (HSMS)

The E37 standard, or HSMS, is designed as a transport protocol for SECS/GEM communications (Li et al., 2022; Zhu et al., 2021). It is based on the Transmission Control Protocol (TCP) and provides a high-speed, reliable, and efficient data exchange mechanism between manufacturing equipment. Since it derives from the TCP/IP protocol, the HSMS protocol employs similar procedures for setting up connections as defined in Request For Comments (RFC) 793, although with minor variations (Rubow, 2003; Sun et al., 2021). While RFC 793 allows both parties to initiate a connection simultaneously, the HSMS protocol limits the connection establishment process. It outlines two distinct modes for establishing connections: Active and Passive. When operating in passive mode, the client waits and listens for incoming connections, whereas in active mode, the client takes the initiative to establish a connection with the server. After establishing a communication link between the host and equipment, the HSMS protocol transmits SECS-II messages in binary format for performing tasks such as controlling recipe changes, monitoring, reporting, and other functions of the semiconductor equipment. The connection between the communicating entities is established for extended periods and maintained throughout the communication process (S. A. Laghari, Manickam, & Karuppayah, 2021). Data exchange occurs between the two entities until one or both are deliberately disconnected for specific reasons such as software upgrades, adding or removing machines, or maintenance. The HSMS message format is illustrated in Figure 2.7, and each of the header fields is described in detail in Table 2.2.

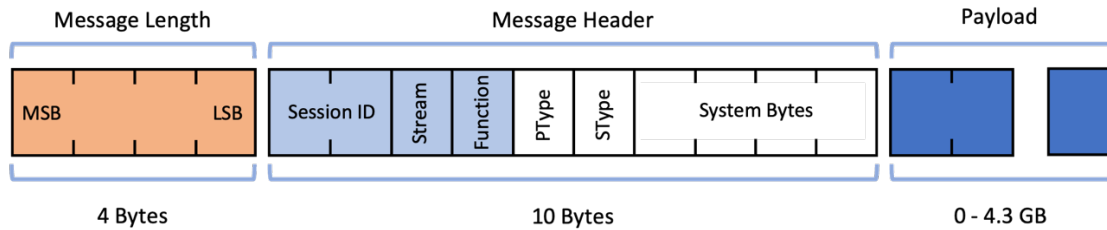


Figure 2.7 HSMS Message Structure

The HSMS message consists of a fixed-length header and a variable-length message data payload. The header is 10 bytes long, and the data payload can range from 0 to 4.3 GB (maximum theoretical size) depending on the message type (Weber, 2016). For example, a data message might have a variable-length data payload that contains the actual data being exchanged, while an acknowledgment message might have an empty data payload.

Table 2.2 HSMS Header Fields Description

Bytes	Field	Description
0-1	SessionID	Provides an association by reference between control messages and subsequent data messages
2	W-Bit & Stream	It is 0 for data messages or a status code for control messages. When SType is 0, it contains W-bit and Stream ID for data messages.
3	Function	It is set to 0 for control messages. When PType is 0, it contains the SECS-II Function ID.
4	PType	Its value is 0 for SECS-II messages.
5	SType	It is set to 0 for data messages. Values 1 to 9 (8 is unused) are for control messages.
6-9	System Bytes	Its value is unique for each transaction.

According to the current HSMS protocol specifications, manufacturing equipment and hosts can only have one active point-to-point communication connection. Therefore, any subsequent connection requests from the equipment will be discarded once connected to the host. Figure 2.8 depicts a scenario where the equipment operates in Passive Mode, waiting for incoming connections on a TCP port (usually 5000).

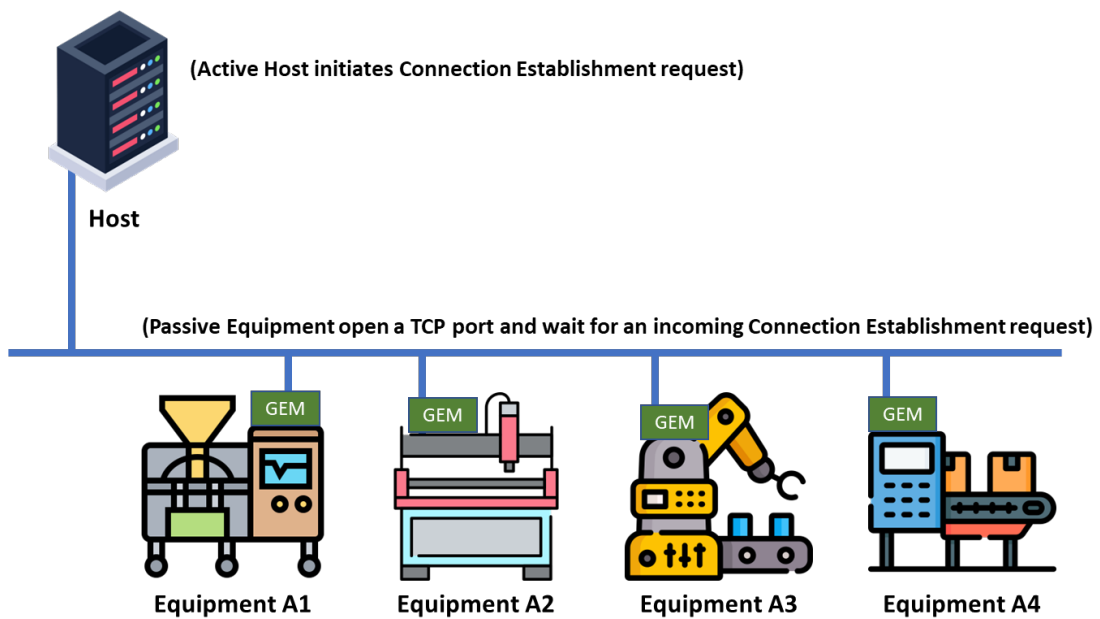


Figure 2.8 Active Host and Passive Equipment Configuration Scenario

The SECS/GEM communication processes are presented in Figure 2.9. To initiate communication, the entity configured in active mode (usually the host) must send a TCP request to the entity configured in passive mode (typically the equipment). Upon receiving a response from the equipment, the host will send an acknowledgment and a TCP connection will be established between the two communicating entities (Figure 2.9(a)). The equipment will then transition into the Connected state and await the establishment of an HSMS connection. At this point, the host will request HSMS link establishment, and the equipment will respond by changing its state from NotSelected to Selected. With the connection established successfully, SECS-II

messages can be exchanged (Figure 2.9 (e)). The process of establishing TCP and HSMS connections is illustrated in Figure 2.9 (a) and Figure 2.9(b).

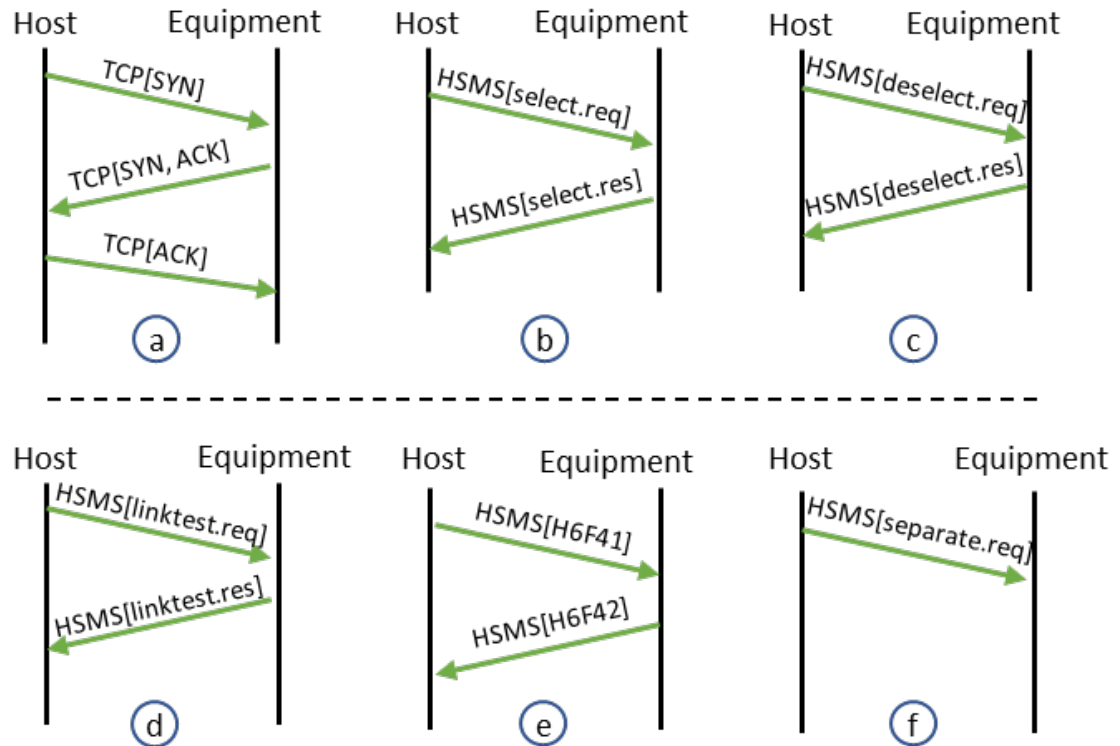


Figure 2.9 SECS/GEM's Connection Establishment, Control, and Data Message Processes

The connections within industrial networks can remain active for several days or weeks. However, there may be instances where two entities do not communicate for a certain period, making it necessary to test connectivity in advance. Figure 2.9(d) shows an example where an inactive period is detected due to a lack of communication between the entities. To prevent this, periodic exchange of control messages "linktest-req" and "linktest-res" occurs based on certain internal timer values, ensuring the connection remains active.

The SECS/GEM specifications outline two methods to terminate a connection: the "deselect-req" or "separate-req" control messages. The main difference between these messages is that "separate-req" immediately cuts off the connection without

waiting for an acknowledgment. An active connection's disconnection phase is shown in Figure 2.9(c) and Figure 2.9(f).

2.2.4 Generic Equipment Model (GEM)

The SEMI E30 standard, also known as GEM, defines a set of specifications for factory equipment. It includes a basic set of requirements that describe generic factory equipment, optional features and functionality, use cases, and scenarios. GEM utilizes a subset of SECS-II messages (Jiao & Peng, 2022), including essential requirements and additional equipment capabilities (Golra & Engel, 2021). The GEM interface is suitable for any factory equipment, regardless of its complexity and size. For example, some equipment may not require recipe management due to the absence of recipes on that equipment, while complex equipment with multiple recipe options necessitates uploading and downloading recipes from the factory host will have a recipe management feature. SECS/GEM is also scalable, accommodating data of various sizes. For instance, simple devices with minimal functionality may publish a few collection events. In contrast, complex equipment may generate massive data, publishing tens of thousands of collection events in a brief period. Despite this difference, both devices can use the same SECS/GEM interface.

The SECS/GEM interface provides comprehensive equipment control and enables factory hosts to monitor all equipment activities. With SECS/GEM, every action on the equipment can be closely monitored, and advanced control policies can be implemented for optimal decision-making. This interface also supports a range of applications, including troubleshooting, predictive maintenance, process monitoring, process controls, material tracking, and equipment utilization, among others. Additionally, SECS/GEM reduces the need for operators to interact with equipment, which minimizes the number of operators required in the factory. Efficient recipe

management systems can help factories reduce material scrap and waste (S. A. Laghari, Manickam, & Karuppayah, 2021).

2.3 SECS/GEM Security Issues

SECS/GEM is vulnerable by design and can be exploited in an Industry 4.0 scenario. This section describes the security issues of SECS/GEM when used in an Industry 4.0 environment. The lack of message authentication and data confidentiality makes SECS/GEM extremely vulnerable to cyber-attacks. The following section highlight the criticality and severity of these issues.

2.3.1 Lack of Authentication Mechanism

After establishing an end-to-end connection between the host and equipment, communication and data exchange may begin. However, to prevent potential damage to reputation and or disruptions in communication, it is crucial for the devices involved to be aware of who they are interacting with and for the interaction to be trustworthy. Authentication and identification of communication devices are necessary for these critical systems (Al-Shareeda et al., 2022). Unfortunately, SECS/GEM lacks an authentication mechanism, allowing threat actors to connect and exchange data with authorized industrial machines without permission (S. A. Laghari, Manickam, & Karuppayah, 2021). These attackers may also capture and forge a bogus message from a legitimate host/equipment and send a “deselect-req” message to terminate the communication link using the SECS/GEM message with SType=9. As a result, the communications may be interrupted without the host ever realizing that the connection was terminated (S. A. Laghari, Manickam, Karuppayah, et al., 2021). Figure 2.10 illustrates how such an attack on SECS/GEM communications is carried out.

Typically, the attacker scans the network to passively identify nodes listening on port 5000, the port used for SECS/GEM communications. The perpetrator then captures and monitors the network traffic, specifically focusing on the most recent response message sent by the targeted device. It should be noted that the probability of a successful attack on response messages is higher than on request messages, as it usually takes some time to request another set of data in quick succession (S. A. Laghari, Manickam, Karuppayah, et al., 2021).

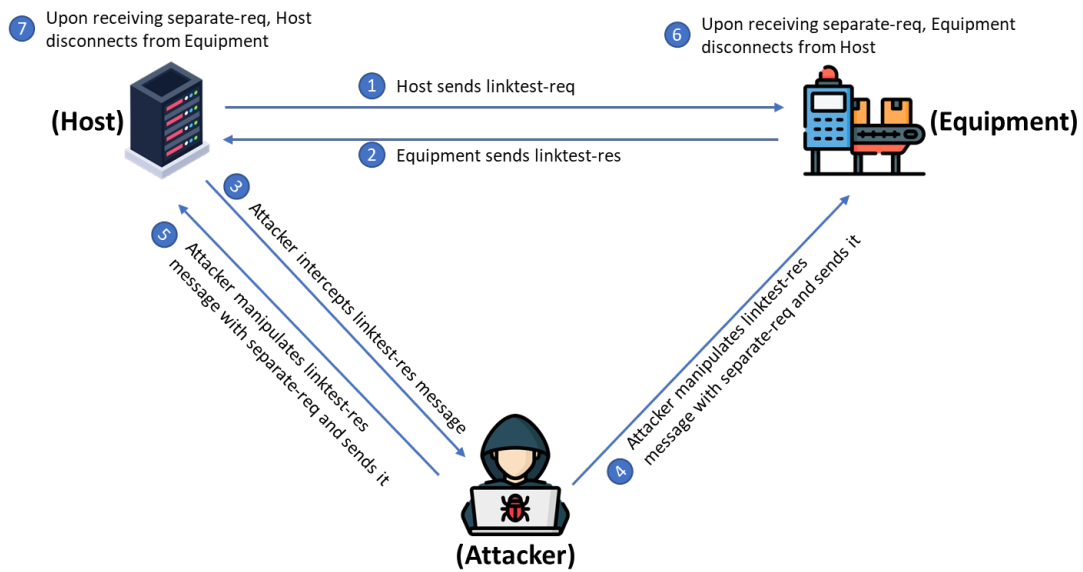


Figure 2.10 Attacker Intercepting, Manipulating and Attacking SECS/GEM Communications

2.3.2 Lack of Message Data Confidentiality

Although SECS/GEM offers several advantages, such as real-time communication, equipment control, and data collection, it also lacks data confidentiality, leading to several issues (S. U. A. Laghari et al., 2023). One of the main issues is the potential for sensitive information to be accessed or intercepted by unauthorized parties, including competitors, which can lead to significant financial and intellectual property losses (Chhetri et al., 2017). This sensitive data can be proprietary process recipes and manufacturing data. If this information falls into the wrong hands, it can lead to severe financial losses for the affected company. For example, it can result

in counterfeit products, supply chain disruptions, and reputational damage. Data breaches can also cause regulatory compliance violations, damage a company's reputation, and cause loss of customer trust (Mohammed, 2021).

Figure 2.11 presents the Wireshark capture of a SECS/GEM message transaction. It can be seen that the binary encoded data is presented in a readable format in Wireshark, including the header and the data in the payload.

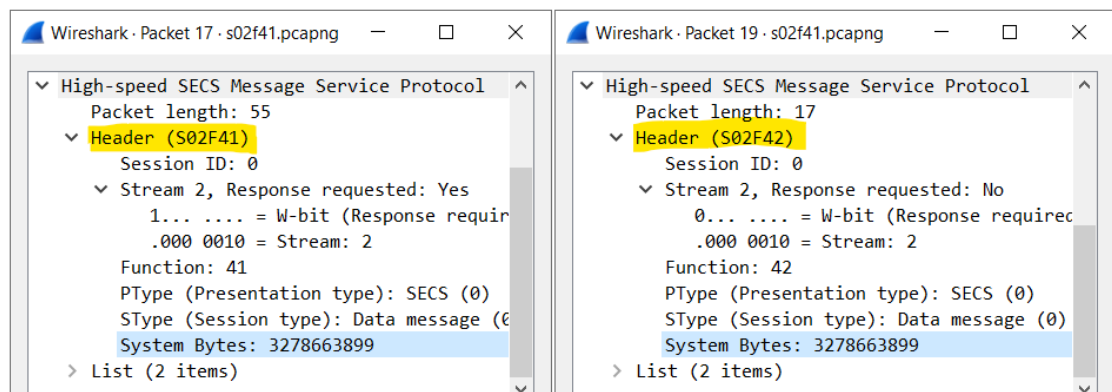


Figure 2.11 Wireshark Capture of SECS/GEM Message Transaction

To sum up, the absence of data confidentiality in SECS/GEM poses a significant challenge for semiconductor manufacturers using equipment that communicate using SECS/GEM. The potential hazards, including data breaches, theft of intellectual property, data integrity problems, and supply chain security issues, demand a robust security measure, like encryption, to safeguard sensitive information and adhere to data privacy laws.

2.4 Securing Data in Transit: Encryption for Network Communications

Encryption in the context of cryptography is a mechanism for ensuring confidentiality (Jyothi et al., 2020). Since data may be visible on the network, sensitive information such as passwords and private information may be accessible to potential interceptors. Encryption is a method for securing data, mathematically encoding it so

only those with the correct key can decode it. Encryption processes data using an algorithm that renders the original information unreadable to all but authorized parties. Encryption is essential in a digitally connected world to keep privileged information, messages, and financial transactions secure and private.

One of the strengths of encryption is determined by the length of the encryption security key. In the final decades of the 20th century, software developers used 40-bit encryption, a key with 2^{40} possible variations, or 56-bit encryption with 2^{56} possible variations. However, by the end of the century, brute-force attacks enabled attackers to crack these keys. Thus, 128-bit encryption became the standard.

The Advanced Encryption Standard (AES) is a data encryption algorithm developed by the National Institute of Standards and Technology (NIST) of the United States in 2001. AES employs a block size of 128 bits and key lengths of 128, 192, and 256 bits (Dworkin et al., 2001; Kuswanto, 2021; Muhammad Abdullah, 2017). AES employs an algorithm with symmetric keys. A symmetric key indicates that the same key is utilized for encryption and decryption. Asymmetric-key algorithms' encryption and decryption processes use different keys (Liestyowati, 2020). 128-bit encryption is the norm today, but most banks, militaries, and governments employ 256-bit encryption (Pathak et al., 2018).

2.4.1 Symmetric-Key Encryption

Symmetric Key Encryption is a type of encryption that uses a single key for data encryption and decryption (Wong, 2021). This key, known as a symmetric key, is shared between the two parties for data transfer and communications over a network. This type of encryption is typically used for fast and efficient encryption and decryption (Ekta & Parashu, 2017).

Symmetric Key Encryption is used for various purposes, from protecting data during transmission over networks to encrypting data to be written to a storage media. It is often used for secure file transfer protocols such as SFTP, as well as for secure storage of data. It is also commonly used in virtual private networks (VPNs) and in applications such as voice over IP (VoIP) and instant messaging (Schatz et al., 2023).

The main strength of symmetric key encryption is its speed, as it can encrypt and decrypt data much more quickly than other types of encryption. However, it is also vulnerable to brute force attacks, when attackers try to guess the key by trying different combinations (Yingbing & Yongzhen, 2014). As a result, when using symmetric key encryption, it is important to choose a key that is long and complex enough to make it difficult to guess. It is also important to ensure the key is kept secure and only shared with trusted parties. Examples of symmetric key encryption algorithms include Advanced Encryption Standard (AES), Data Encryption Standard (DES), Triple Data Encryption Standard (3DES), etc. Since these encryption algorithms can only provide data confidentiality, they are not chosen for this study. Figure 2.12 illustrates a high-level view of how Symmetric Key encryption works.

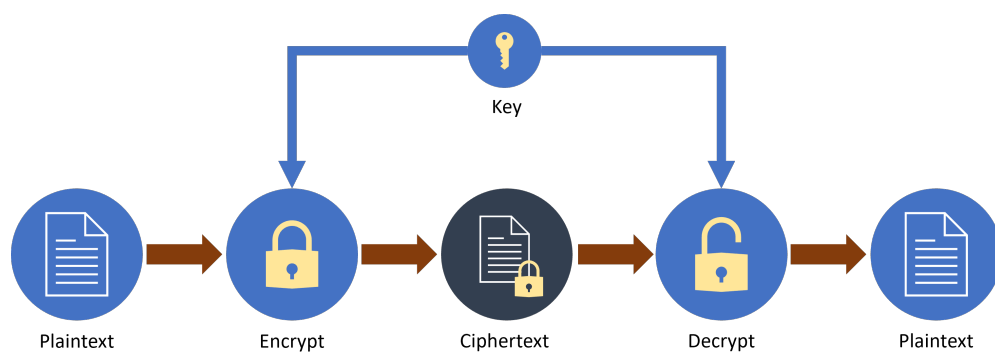


Figure 2.12 High-level View of Symmetric Key Encryption

2.4.2 Asymmetric-Key Encryption

Asymmetric key encryption, also known as public-key cryptography, is a type of encryption that uses two different keys for encryption and decryption (Halak et al.,

2022). One key is kept private, while the other is shared with the public. This type of encryption protects data from unauthorized access. It is widely used in secure communication protocols such as Secure Sockets Layer/Transport Layer Security (SSL/TLS) and Secure Shell (SSH) (Baka et al., 2020).

Asymmetric key encryption is based on the concept of public and private keys. The public key is used to encrypt the data, while the private key is used to decrypt it. To send a secure message, the sender encrypts the data using the recipient's public key and sends it over the network. The recipient can then decrypt the data using their private key. This way, only the recipient can read the message, making it impossible for anyone else to access the data.

Asymmetric key encryption is used in various applications, such as digital signatures, secure web browsing, and secure email (Halak et al., 2022). It is also used in secure payment processing, data encryption, and secure file sharing (Bojjagani et al., 2021). Asymmetric key encryption is considered one of the most secure encryption methods available. Many organizations and individuals use it to protect their data from unauthorized access. The Rivest–Shamir–Adleman (RSA) and Elliptic-curve cryptography (ECC) are examples of asymmetric key encryption algorithms. Similar to the symmetric key encryption algorithms, they only provide data confidentiality. Hence, these encryption algorithms are not used in this study. Figure 2.13 illustrates a high-level view of how Asymmetric Key encryption works.

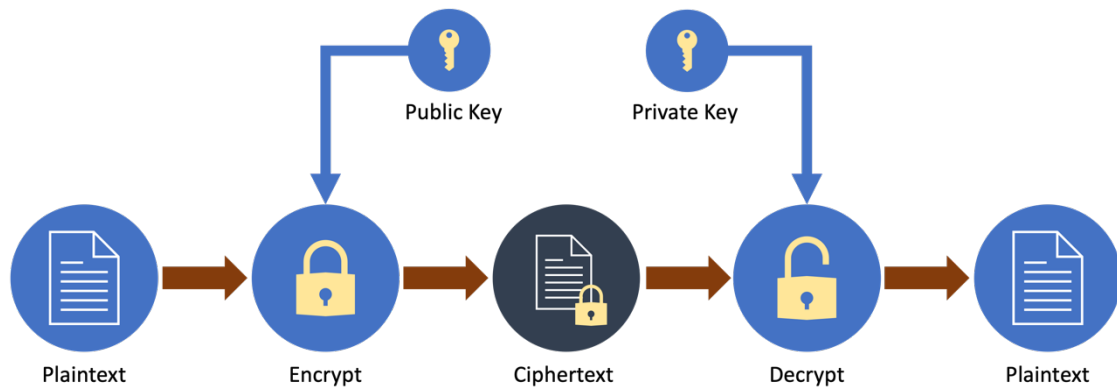


Figure 2.13 High-level View of Asymmetric Key Encryption

2.4.3 Authenticated Encryption

Authenticated Encryption (AE) and Authenticated Encryption with Associated Data (AEAD) are encryption techniques that provide data confidentiality and authentication (Pirzada et al., 2020). It is a form of encryption that also verifies the authenticity of the encrypted data, meaning that the data can be both read and authenticated as having come from a legitimate source. Thus, unauthorized entities cannot tamper with the data and do not have access to the actual unencrypted data (Jimale et al., 2022).

Authenticated encryption can be used in a variety of applications. It is often used in secure communication between two parties, such as in a virtual private network (VPN). It can also protect data stored in databases or other digital storage systems. In addition to protecting data, authenticated encryption can also be used to verify the sender's identity. It is crucial in applications such as electronic banking and ecommerce.

Authenticated encryption is a more secure form of encryption than traditional encryption because it provides both data confidentiality and authentication (Wong, 2021). It is also more efficient than conventional encryption methods, as it can encrypt and authenticate data simultaneously. As a result, authenticated encryption is becoming increasingly popular for various applications, from secure communication to data

storage. AES with Galois/Counter Mode (AES-GCM), AES - Counter with CBC-MAC (AES-CCM) and ChaCha20-Poly1305 are few examples of authenticated encryption algorithms. Figure 2.14 illustrates a high-level view of how Authenticated encryption works.

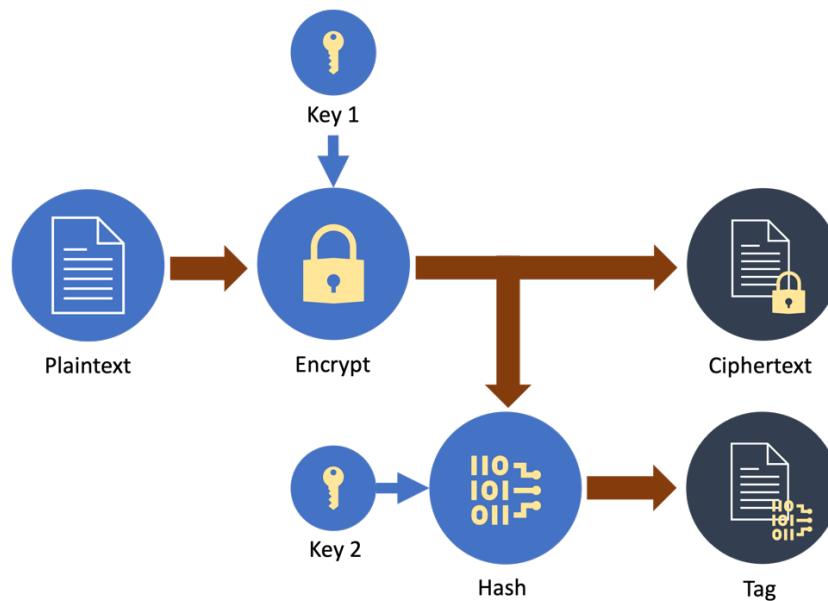


Figure 2.14 High-level View of Authenticated Encryption

Table 2.3 compares the various features of symmetric key encryption, asymmetric key encryption and authenticated encryption. It can be seen from this comparison that authenticated encryption provides all the features required to accomplish the objectives of this study.

Table 2.4 compares popular authenticated encryption algorithms. It can be deduced that AES-GCM is more widely adopted compared to other popular algorithms. AES-GCM is would also be able to provide better speeds, up to 10 Gbps, with low-cost hardware accelerators are available (Abdellatif et al., 2017). Therefore, AES-GCM is chosen as the preferred authenticated encryption algorithm for this study.

Table 2.3 Comparison of Features Between Different Encryption Techniques

Feature	Symmetric Key Encryption	Asymmetric Key Encryption	Authenticated Encryption
Key Types	Single key for both encryption and decryption.	Two keys: Public key for encryption, private key for decryption.	Usually involves both symmetric and asymmetric keys for encryption.
Computational Complexity	Generally faster and more efficient.	Slower compared to symmetric encryption.	Slightly slower due to additional processes for authentication.
Use Cases	• Bulk data encryption • Communication over secure channels.	• Secure key exchange (e.g., RSA) • Digital signatures for authentication.	• Secure communication with data integrity. • Transmitting sensitive data with protection against tampering.
Security Strength	Dependent on key length. Longer keys provide higher security.	Dependent on the difficulty of factoring large numbers.	Combines encryption and authentication for a higher level of security.
Examples	AES, DES, 3DES	RSA, ECC	AES-GCM, AES-CCM, ChaCha20-Poly1305

Table 2.4 Comparison of Popular Authenticated Encryption Algorithms

Feature	AES-GCM	AES-CCM	ChaCha20-Poly1305
Underlying Cipher	AES	AES	ChaCha20
Security Level	128-bit, 192-bit, 256-bit	128-bit, 256-bit	256-bit
Hardware Support	Widely available (Bühler et al., 2022)	Widely available	Less widespread
Performance	Generally fast. Better performance with hardware acceleration (Skuballa et al., 2021)	Generally fast, but slower than AES-GCM	Generally fast. Slower compared to AES-GCM with hardware acceleration (Bühler et al., 2022; Mosnáček, 2018)

2.4.4 Overview of AES-GCM

Advanced Encryption Standard (AES) with Galois/Counter Mode (GCM) is a symmetric encryption algorithm that provides both confidentiality and authenticity of the data. AES-GCM is widely used for secure communication over the Internet, and it is considered one of the most secure encryption algorithms available today (Menezes & Stebila, 2021).

AES-GCM combines AES, a block cipher that operates on fixed-length blocks of data, and GCM, a mode of operation that provides authentication and confidentiality for the data. AES is used for encryption and decryption, while GCM is used for authentication. GCM is based on the Galois Field (GF) multiplication, a mathematical operation for generating the authentication tag (Dworkin, 2007). The authentication tag ensures that the data is not tampered with during transmission.

AES-GCM is fast and efficient, making it ideal for applications requiring high-speed encryption and decryption, such as network communication, cloud storage, and consumer file storage solutions. It is also resistant to brute force, side channel, and known-plaintext attacks. Overall, AES-GCM is a strong and widely adopted encryption standard (Koteshwara et al., 2017), that provides both confidentiality and authenticity of the data, making it a popular choice for secure communication.

The primary objective of this study is to achieve confidentiality and authentication in data communication. To accomplish this, AES-GCM is a suitable algorithm to achieve the desired objectives. AES-GCM is a highly secure and efficient encryption algorithm that offers confidentiality and authentication. By utilizing AES-GCM, this study is expected to ensure that the data transmitted in SECS/GEM communications are authenticated and remain confidential, thus providing secure communication.

2.5 Related Work

Protecting intellectual properties and sensitive information in the production environment is vital for companies to maintain competitiveness in the Industry 4.0 landscape. Historically, the manufacturing sector has prioritized safeguarding enterprise perimeters. The physical location of manufacturing equipment and networks was protected to prevent unauthorized access to the production network. Consequently, cybersecurity was not a primary design goal during the development of protocols, as they were primarily designed for use in closed, air-gapped, and trusted industrial networks. However, Industry 4.0 requires interconnectivity with industrial networks to access real-time equipment/machine data, necessitating robust cybersecurity protocols to safeguard against cyber-attacks.

In recent years, Industry 4.0 has emerged as a significant driving force for the manufacturing industry, bringing the M2M protocols to the forefront of attention among researchers and experts. Unfortunately, these protocols were designed primarily with high performance in mind and, thus, lack adequate security measures, leaving them vulnerable to cyber-attacks. Researchers are working on solutions to make these protocols secure for communication in the Industry 4.0 environment. The following subsections briefly summarize the security features offered by these solutions.

The significance of cybersecurity issues in the manufacturing sector has recently been brought to the attention of researchers. As a result, research has been conducted to recommend design and propose security measures for Industry 4.0 and IIoT (Yu & Guo, 2019). The industrial network's smooth functioning relies heavily on network devices' participation. A single unauthorized and malicious node can halt the entire production line and cause catastrophic consequences. Therefore, it is imperative for the equipment

and machinery involved in the industrial environment to establish a reliable and secure connection and only communicate with trusted and authorized devices. The initial defense against cyber-attacks on industrial networks is to make it challenging or impossible for adversaries to establish unauthenticated communication links with legitimate manufacturing equipment.

No security features are defined in the SECS/GEM specifications to identify unauthorized messages in SECS/GEM communications. The transport protocol for SECS/GEM, HSMS, outlines the equipment's states and facilitates real-time control and monitoring of operations by the host. With SECS/GEM's remote-control capabilities, operators can configure and manage equipment effectively. Nonetheless, if the communication channel is breached, malicious actors can misuse these features to interfere with the manufacturing processes and disrupt communication. Following are the studies addressing SECS/GEM's security concerns.

The SECS/GEMsec security mechanism was developed to ensure message authentication for SECS/GEM communications. It combines asymmetric key encryption techniques with hashing algorithms to authenticate messages. The development of this mechanism had four main objectives. The first objective was to prevent the alteration of SECS/GEM messages during transit. The SECS/GEMsec mechanism utilizes SHA-256 to maintain message integrity and achieves this objective (S. U. A. Laghari, Manickam, Al-Ani, et al., 2021a). SHA-256 was chosen due to its unbroken status, reasonable security, and fast performance. To transmit a message, SECS/GEMsec computes the hash value using SHA-256 and further processes it to generate the message signature.

The second objective of the SECS/GEMsec security mechanism is to prevent unauthorized devices from communicating with legitimate factory equipment by implementing a sophisticated authentication mechanism. Digital signature encryption encrypts the message hash on the sender side and decrypts it on the receiver side during verification, thus achieving the second objective. Asymmetric-key cryptography is employed through digital signature algorithms, with the RSA algorithm utilizing a 2048-bit key size to encrypt the hash generated on all SECS/GEM messages. Although a key size of 4096 bits provides greater security than 2048 bits, it doubles the control overhead. The key size of 2048 bits is chosen to reduce the control overhead associated with each message transmitted between a host and equipment. The receiver is preconfigured with the sender's public key to decrypt the message. Without the private key, the message cannot be forged, which enables the receiver to conclusively accept the message as legitimate, transmitted by the authorized node.

The third design objective of the security mechanism is to detect anomalous behavior and prevent unauthorized messages from interfering with communication between the two SECS/GEM devices. The SECS/GEM header utilizes a four-byte field called SystemBytes, which increments monotonically with each request message (S. A. Laghari, Manickam, Karuppayah, et al., 2021). This field serves two critical functions. Firstly, it guarantees that a message is processed only once, as each new SECS/GEM request message will always contain a new SystemBytes value. Secondly, the SystemBytes value in the reply message will match that of the corresponding request message, preventing the acceptance of messages with forged SystemBytes values. Attackers attempting to create new messages using incremented SystemBytes values will be rejected, as the message signature cannot be forged without the private key.

Therefore, the SECS/GEM security mechanism can achieve its third objective by identifying and discarding messages with old, duplicate, or forged SystemBytes values.

SECS/GEMsec's fourth objective is to design a simplistic mechanism that does not alter the existing message structure or packet format. To achieve this, SECS/GEMsec utilizes TCP flow and appends the message signature at the end of the SECS/GEM message without modifying any message fields. Therefore, it eliminates the need for a separate control message to transmit the encrypted hash since the message signature is included in the same TCP payload as the message. To append the signature at the end of the SECS/GEM message, SECS/GEMsec examines the message length value, which can be obtained from the first 4 bytes of a SECS/GEM message header. Thus, the process is simplified and requires no modification to the message structure.

2.6 Discussion

Given the growing emphasis on networked systems and the critical transition to Industry 4.0, the need to protect industrial networks against cyber threats has grown in importance. The safeguarding of assets in this context necessitates strict procedures, particularly when it comes to securing Machine-to-Machine (M2M) connections within industrial networks. Notably, while the widespread SECS/GEM standard is fundamental in industrial communication, it lacks basic security features, making it unsuitable for deployment in the dynamic Industry 4.0 ecosystem. Unfortunately, the academic landscape surrounding this gap remains rather minimal, with few attempts made to address the developing difficulties.

The susceptibility of SECS/GEM communications to intellectual property theft and the compromise of sensitive information, as discussed in Section 2.3.2, emphasizes the need for robust security methods. The fundamental goal of this study is to overcome

the significant issue of developing a security mechanism that assures confidentiality and authentication for SECS/GEM connections.

Despite successfully achieving its design goals, the SECS/GEMsec suffers from problems that make it unusable in live production systems. The following shortcomings have been identified while reviewing the SECS/GEMsec mechanism:

1. **Complex Authentication Process:** SECS/GEMsec employs two pairs of distinct public and private key sets on the communicating entities to ensure authentication and prevent unauthorized messages. However, the process of managing these two pairs of keys for each device can get a bit complicated and can be a hassle.
2. **Cryptography Overhead:** SECS/GEMsec uses SHA-256 cryptographic hash function to create a message digest for verifying that the message has not been tampered with during transit. However, attackers could generate their message digest and insert it into the message to circumvent the system's security. Therefore, the RSA public-key cryptography system is used to encrypt the hash. Unfortunately, using two different methods to achieve the desired security goals results in additional computational overhead, which slows down the overall system performance.
3. **Control Overhead Challenges:** SECS/GEMsec utilizes the RSA algorithm with a key size of 2048 bits, generating 256 bytes of control overhead for each message exchanged between the host and the equipment. This overhead represents the minimum amount of control overhead required. However, SECS/GEMsec advises raising the RSA key size to 4096 bits (512 bytes) to enhance security, increasing the

overhead to 512 bytes. Therefore, excessive control overhead for every message could negatively impact bandwidth consumption.

4. Lack of Confidentiality: The SECS/GEMsec protocol offers authentication and safeguards against unauthorized messages, ensuring their integrity. However, it falls short of providing essential features like confidentiality of data in messages, thus, making the communication vulnerable to passive attacks where attackers can read the messages or obtain sensitive information from them.

Table 2.5 summarizes the strengths and limitations of standard SECS/GEM and SECS/GEMsec.

Table 2.5 Strengths and Limitations of Standard SECS/GEM and SECS/GEMsec

Mechanism	Strengths	Limitations
Standard SECS/GEM (S. A. Laghari et al., 2021)	N/A	• No message authentication • No data confidentiality for messages
SECS/GEMsec (S. U. A. Laghari, Manickam, Al-Ani, et al., 2021a)	• Offers message authentication. • Ascertains message integrity. • Detection and prevention of unauthorized messages	• Using a digital signature-based solution requires managing multiple keys • Using hashing and encryption algorithms to provide message authentication creates additional computational overhead. • Using RSA with a 2048-bit key creates 256 bytes of overhead. • Using RSA with a 4096-bit key creates 512 bytes of overhead

2.7 Chapter Summary

This chapter provided a detailed discussion on the background of the SECS/GEM communication protocol, its underlying SEMI standards (i.e., SECS-I, SECS-II, HSMS, and GEM), and fundamental concepts associated with this research.

In addition, the importance of security in SECS/GEM communications in industrial networks was discussed thoroughly. In addition, the connection management processes, message structure, message encoding, and message formatting were all covered in depth in this chapter. The security issues associated with standard SECS/GEM were identified. Encryption techniques were explored with a focus on the AES-GCM algorithm. The SECS/GEMsec security mechanism was also reviewed, and its limitations were identified. Chapter 3 proposes a novel security mechanism to achieve the objectives of this study while addressing the issues identified with the related work.

CHAPTER 3

METHODOLOGY

This chapter explains in depth the proposed SECS/GEM security mechanism referred to as SECS/GEM with Authenticated Encryption (SECS/GEM-AE). Control information used by SECS/GEM-AE is carried with every message. The control information is a core part of SECS/GEM-AE as it is essential to authenticate messages and check if they have not been changed during transmission. The control information is piggybacked with every SECS/GEM message so as not to change the original message structure of SECS/GEM messages.

The proposed mechanism consists of three stages: Stage 1, Authenticated Encryption; Stage 2, Control Information Generation; and Stage 3, Decryption and Authentication. In the first stage, the message is encrypted, and authentication information is generated for verification at the receiver's side. This stage accomplishes the first and second objectives of the study. In the second stage, control information is attached to the message. In the third stage, the receiver utilizes the control information to decrypt and authenticate the message. Detailed discussions of these stages are provided in the following sub-sections.

The remainder of this chapter is structured as follows. Section 3.1 discusses the proposed mechanism's methodology, assumptions, and threat model. The proposed mechanism, its components, control information, and underlying mechanisms are presented in Section 3.2. The workflow of the proposed mechanism is discussed in Section 3.3; Section 3.4 explains the assessment methods and measures. Section 3.5 summarizes the entire chapter.

3.1 Developing the SECS/GEM-AE Mechanism: Methodology and Approach

SECS/GEM-AE intends to provide a viable security mechanism capable of providing data confidentiality and ensuring the authentication of SECS/GEM messages. Data confidentiality ensures unauthorized devices can no longer read the SECS/GEM message payload. Only legitimate messages are processed through message authentication, and any altered messages will automatically be dropped.

3.1.1 Underlying Assumptions for the Proposed Mechanism

Several assumptions are made for developing a comprehensive security mechanism designed to fulfill this study's research objectives and validate the anticipated results. The following assumptions are made.

- As the SECS-I protocol is obsolete and only survives on older machines, this research is confined to using the HSMS protocol.
- In order to assess the functionality and performance, all industrial network participants (hosts and equipment) are anticipated to be set up with the proposed mechanism i.e., SECS/GEM-AE mechanism. This is important because the message contents are encrypted, due to which the recipient cannot understand and would discard message.
- Since the proposed mechanism uses symmetric encryption algorithm, which requires a secret key to encrypt and decrypt the message before processing it. The sender and the receiver must possess the same secret key to decrypt and verify the authenticity of a received message. This requirement is met by the SECS/GEM-AE, where the secret keys are preloaded on the communicating devices during initialization. Thus, in this study, it is assumed that participating nodes are preconfigured with the required keys.

3.1.2 Assessing Security Risks: The Threat Model for the SECS/GEM-AE

The threat model is described as follows to present a comprehensive security mechanism that achieves the objectives specified in Section 1.4:

1. The attacker can monitor network traffic and knows the network port used by the host and equipment.
2. The attacker has knowledge of SECS/GEM messages (streams and functions) and binary encoding, enabling the attacker to read messages and passively collect sensitive data.
3. The attacker can inspect messages between the two interacting entities and send crafted or modified messages with malicious intent.

3.1.3 Design Goal of the SECS/GEM-AE Mechanism

Due to the lack of security features in standard SECS/GEM and the shortcomings of existing SECS/GEM security mechanisms, a new security mechanism that meets the research objectives provided in Chapter 1 is required. The literature review reveals various issues in SECS/GEMsec, designed to protect SECS/GEM communications. Based on the literature review in Chapter 2, the criteria for developing a new security mechanism for SECS/GEM communications in the industrial network are as follows:

1. **Lightweight:** The lightweight requirement can be met by simplifying the existing data confidentiality and message authentication procedures. As SECS/GEMsec, which uses SHA-256 and RSA to ensure authentication, requires much processing time, an authenticated encryption function can reduce its complexity. Thus, the mechanism is

much lighter and more efficient as it removes the need for RSA and the manual configuration of three separate keys for SHA-256 and RSA on each computer of the industrial network.

2. **Multi-featured Solution:** The SECS/GEMsec mechanism only provides authentication. A robust system should provide multiple security features in one solution; the proposed mechanism combines multiple features like data confidentiality and message authentication.

3.2 Design of the Proposed SECS/GEM-AE Mechanism

The proposed SECS/GEM-AE mechanism protects industrial network communications to achieve the research objectives in Chapter 1 (data confidentiality and authentication). This section describes the mechanism's primary stages, processes, and components. Figure 3.1 shows the proposed mechanism's operating layer in SECS/GEM architecture.

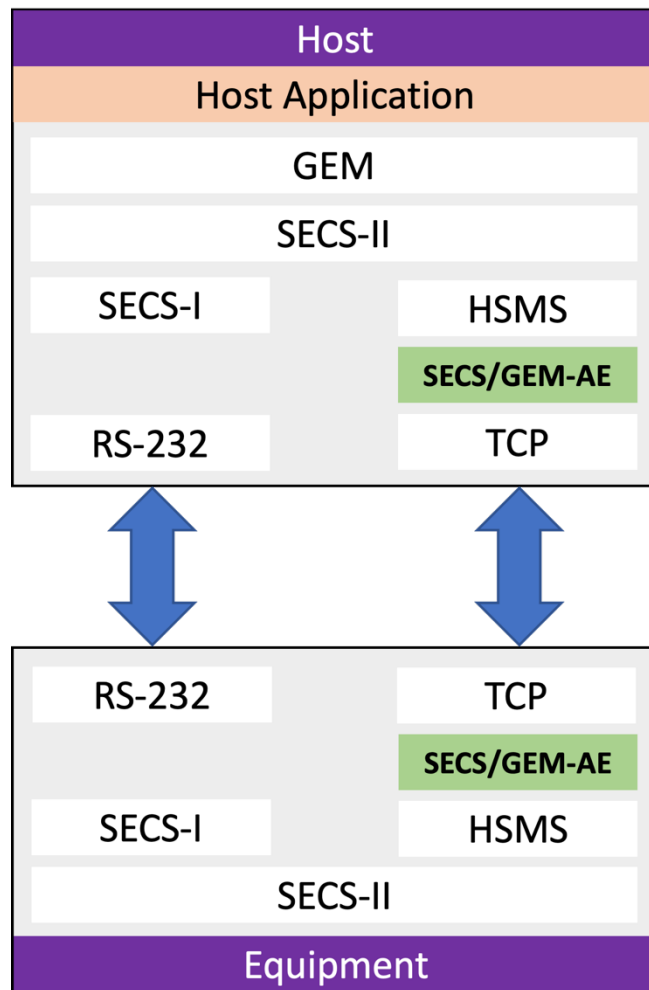


Figure 3.1 Proposed Mechanism's Operating Layer in SECS/GEM Architecture

The proposed SECS/GEM-AE mechanism has three stages. The first is the authenticated encryption mechanism. In this stage, the SECS/GEM message is fed into the authenticated encryption mechanism to generate the encrypted data and tag. Stage 2 is control information generation. In this stage, the control information is generated to authenticate communications. Stage 3 is achieving message authentication. This stage determines if messages are authentic or illegitimate. Figure 3.2 displays the SECS/GEM-AE security mechanism's stages.

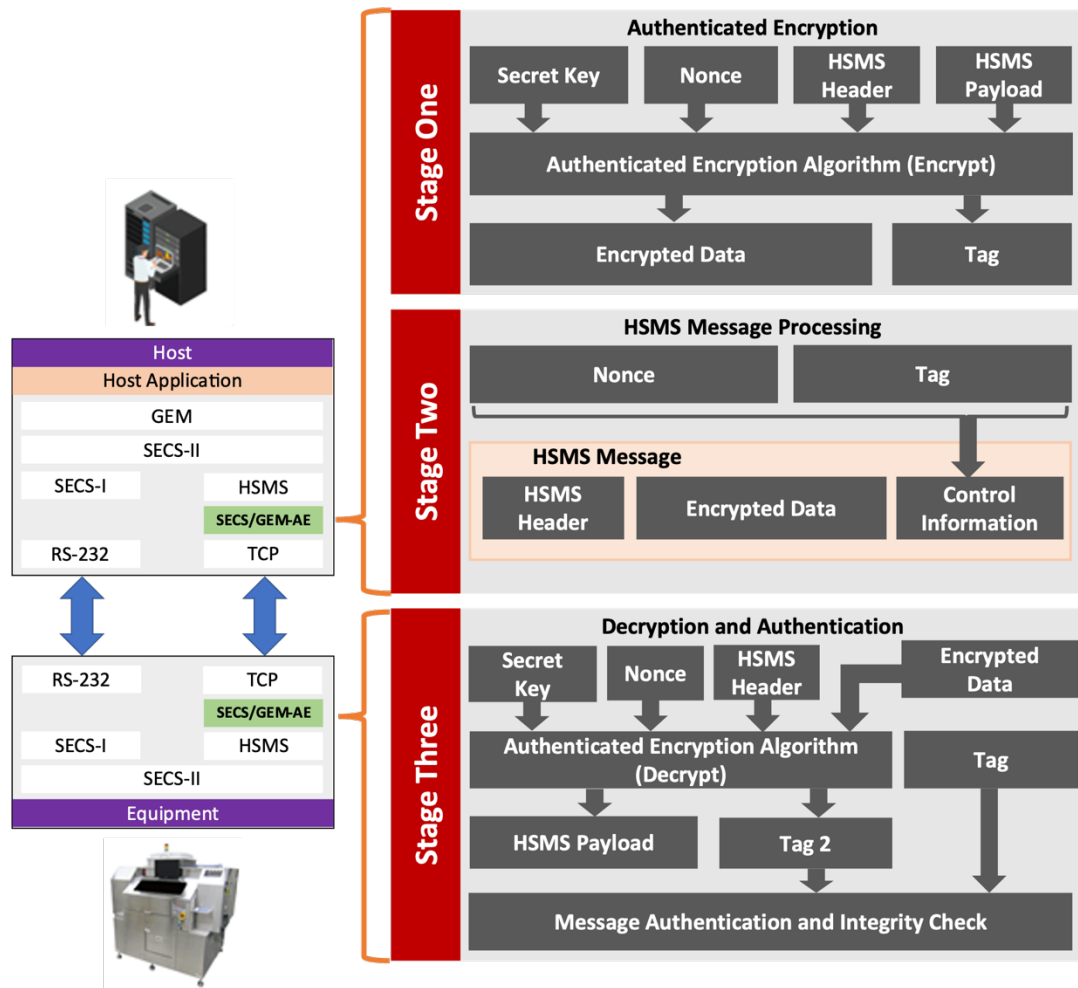


Figure 3.2 SECS/GEM-AE Mechanism Stages

3.2.1 Authenticated Encryption (Stage 1)

This stage aims to achieve the first objective of this research, which is to exchange encrypted SECS/GEM messages rather than plaintext messages to protect sensitive information within SECS/GEM communications in the manufacturing environment. A desired authentication encryption scheme is used to achieve this goal. This research utilizes the AES-GCM algorithm to encrypt the messages and produce tags to verify and authenticate the messages.

In the first stage, the secret key, pseudorandom nonce, plaintext data to be encrypted, and associated data to be authenticated are given as input to the encryption algorithm. In this case, the message payload is encrypted while the message header is

used as associated data which is only authenticated and not encrypted. The encrypted message payload and the verification tag are generated as output. The encrypted message payload replaces the message payload. Figure 3.3 depicts the Host, Equipment, and SECS/GEM-AE configuration used. The configuration includes IP address, port, mode of operation, algorithm used, and secret keys. The secret keys (K1, K2) used by the host and equipment are installed in the host and the respective equipment. The host maintains a list of equipment (E1, E2) and maps the respective key (K1, K2) to each equipment.

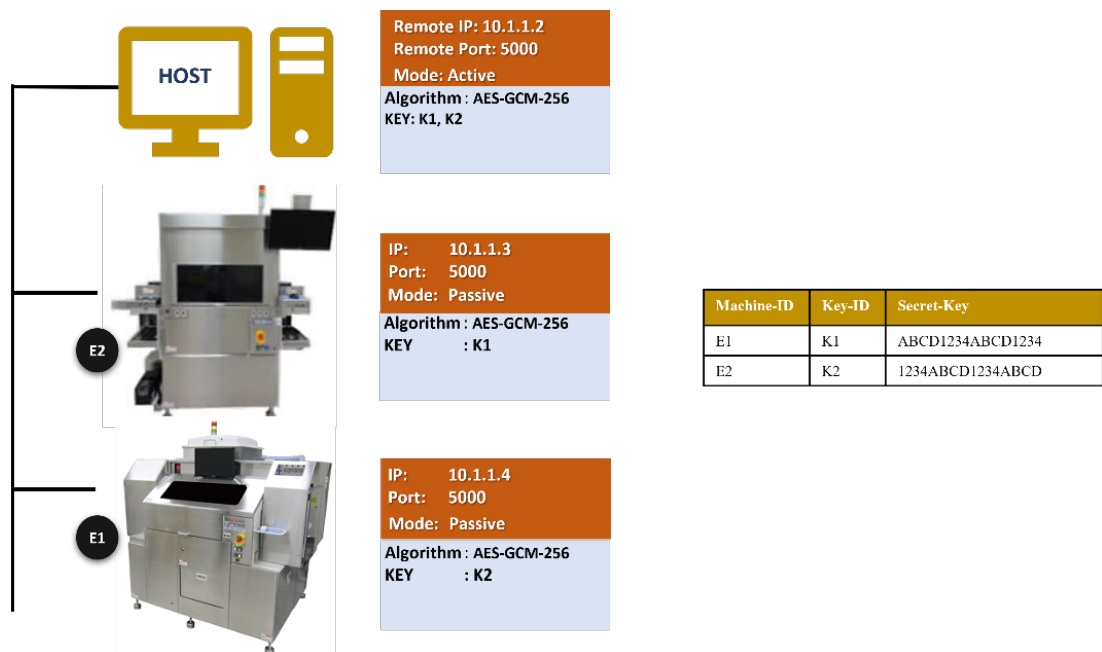


Figure 3.3 Host and Equipment with SECS/GEM-AE Configuration

Thus, the first and second objectives of this study are achieved. The verification tag is essentially a hash of the message header and payload. This tag will be used at the receiver's side to authenticate the message and verify its integrity. Thus, the first objective of this study is achieved. The message payload is also encrypted, making the data in the message payload confidential and therefore achieving the second objective of this study.

3.2.2 Message Processing and Control Information (Stage 2)

The control information is at the core of the proposed mechanism. It includes all necessary information to resolve security flaws of standard SECS/GEM communications and issues in existing security mechanisms. The control information comprises Nonce and Tag that the receiver requires to decrypt the message and ensure message authenticity. The structure of the control information is depicted in Figure 3.4.

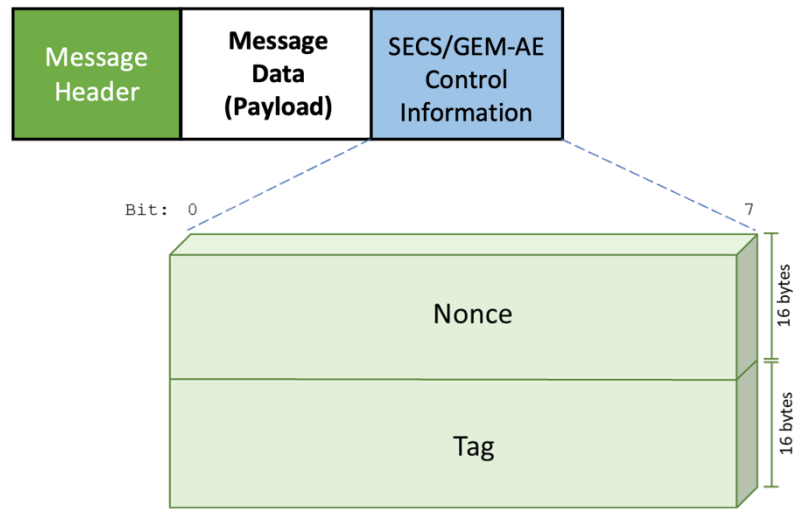


Figure 3.4 Control Information Structure

- **Nonce Field:** This 16-byte field contains the pseudorandom nonce used by the authenticated encryption used with SECS/GEM-AE. This nonce value is required at the receiver to decrypt the message payload.
- **Tag field:** This 16-byte header field contains the Tag generated by the encryption scheme. The receiver uses the Tag to verify the authenticity of the message.

3.2.3 Decryption and Message Authentication (Stage 3)

The receiver must decrypt and verify the integrity of the message payload for data messages before accepting them. It is important to note that this process does not apply to control messages since they do not contain any payload. This stage is about

decrypting the message and authenticating the message. The first step is to extract the information contained in the control information, i.e., nonce and tag. The nonce is used as an input for the decryption process. The tag is used to verify the message's authenticity and integrity at the end of the decryption process.

The secret key, nonce, encrypted payload, and message header (as associated data) are input for the decryption algorithm. The encrypted payload is decrypted to its original form, and at the end of the decryption process, the receiver regenerates a tag (Tag2) using the same key. Suppose the generated tag (Tag2) matches the tag that arrived with the received message. In that case, the message is considered authentic, ensuring no changes have occurred during transit and message integrity is maintained. However, the message is dropped if there is a mismatch between the tags.

3.3 SECS/GEM-AE Workflow

The SECS/GEM request messages are processed at the sender and the commands are packaged in the specified SECS/GEM structure. Once this stage is complete, the proposed mechanism comes in the action and intervene and takes this data as input before the data is transferred to the destination. Once the input is available to the sender, the rest of the details of the workflow of the proposed mechanism are listed below:

1. Extract SECS/GEM Header: SECS/GEM-AE only checks the SECS/GEM header's integrity while encrypting the message payload. The sender needs to extract the SECS/GEM header to be used as associated data during the encryption process.

2. Extract SECS/GEM Payload: The SECS/GEM messages are of two types: control messages and data messages. The control messages are usually comprised of the header only; thus, these messages must be authenticated. In contrast, the data messages usually have a payload, which must be encrypted. Thus, the sender must extract the payload and process it differently from the header.
3. Retrieve Secret Key: Dependent upon the algorithm selected, the secret size may vary; thus, the right key must be selected for the encryption and authentication process. The host usually connects with more than one piece of equipment at a time; hence, the right key must be chosen based on the communicating equipment.
4. Generate Random Nonce: The pseudo-random number (nonce) used to initialize the encryption algorithm is generated next.
5. Encryption Process: The process of encryption involves using an encryption algorithm, such as AES-GCM with a preferred key size, to encrypt the message payload data. The message header is used as associated data input, which is only authenticated. The Secret Key and Nonce are used in this process.
6. Authentication Tag: Upon the completion of the encryption process, the encrypted payload and a tag (Tag1) will be generated by the given algorithm, which the receiver requires to authenticate the received message.

7. Replace Message Payload: Once the header and data contained in the SECS/GEM message are processed, the SECS/GEM message payload is replaced with the encrypted payload.
8. Append Control Information: The nonce and tag obtained in steps 5 and 7 are appended to the SECS/GEM message as control information so that the receiver can use the nonce to decrypt and verify the message. The SECS/GEM header's first 4 bytes contain the length of the message. The proposed mechanism uses the value stored in this field and places the control information at length + 1 location. This way, the control information is appended to the messages and is carried out by the message itself as piggybacked.
9. Send SECS/GEM message: The control information is piggybacked with the message; thus, the message data confidentiality and authentication are achievable without using separate control messages exchanged between the sender and the receiver.
10. Check Control Information: Upon receiving the message, it is only processed if it has control information; otherwise, the message is discarded.
11. Processing Received Message: Upon receiving a SECS/GEM-AE message, the receiver must first extract the SECS/GEM header, encrypted payload, and control information contained in the message. The nonce, tag, header, and encrypted payload are extracted from the message and saved in temporary variables.

12. Retrieve Secret Key: Dependent upon the key size configured, the secret key may vary; thus, the right key must be selected for the decryption and authentication process.
13. Decryption Process: Decrypt encrypted payload with the decryption algorithm (e.g., AES-GCM with preferred key size), Secret Key, and Nonce using Header as input for associated data.
14. Authentication Tag: Upon completion of the decryption process, the original message payload and a tag (Tag2) will be generated by the given algorithm. The receiver requires this tag to authenticate the received message.
15. Message Authentication: Verify tag (Tag2) generated from decryption with the tag (Tag1) extracted from the control information. If both tags match, the message is valid, or else, the message is invalid.
16. Accept/Reject Message: If the message is valid, the encrypted payload is replaced with the original message payload (obtained after decryption). The control information is removed, and the message is processed. If the message is invalid, the message is dropped.

The proposed mechanism uses authenticated encryption. The message payload is encrypted, while the message header is used as associated data (authenticate only). For every message, a random 16-byte nonce is generated for encryption. Once encryption is complete, the encrypted message payload and a tag (Tag1) are generated. The receiver uses this tag to verify the integrity of the message. The message payload

is replaced with the encrypted payload. The nonce and the tag are attached at the end of the message as control information. The message is then sent to the receiver.

The receiver checks the message for the control information and discards it if it is absent. The nonce and tag (Tag1) are extracted from the control information. The receiver decrypts the encrypted payload using the message header as associated data (authenticate only). The nonce present in the control information is also used as input for decryption. The decryption yields the original message payload and a tag (Tag2). If the tag from the message's control information (tag1) matches the tag generated after decryption (Tag2), the message is considered authentic and valid. If the message is valid, the control information is removed from the message, and the encrypted payload is replaced with the original message payload and accepted. If the message is not valid, it is discarded. Figure 3.5 depicts the workflow of the proposed mechanism.

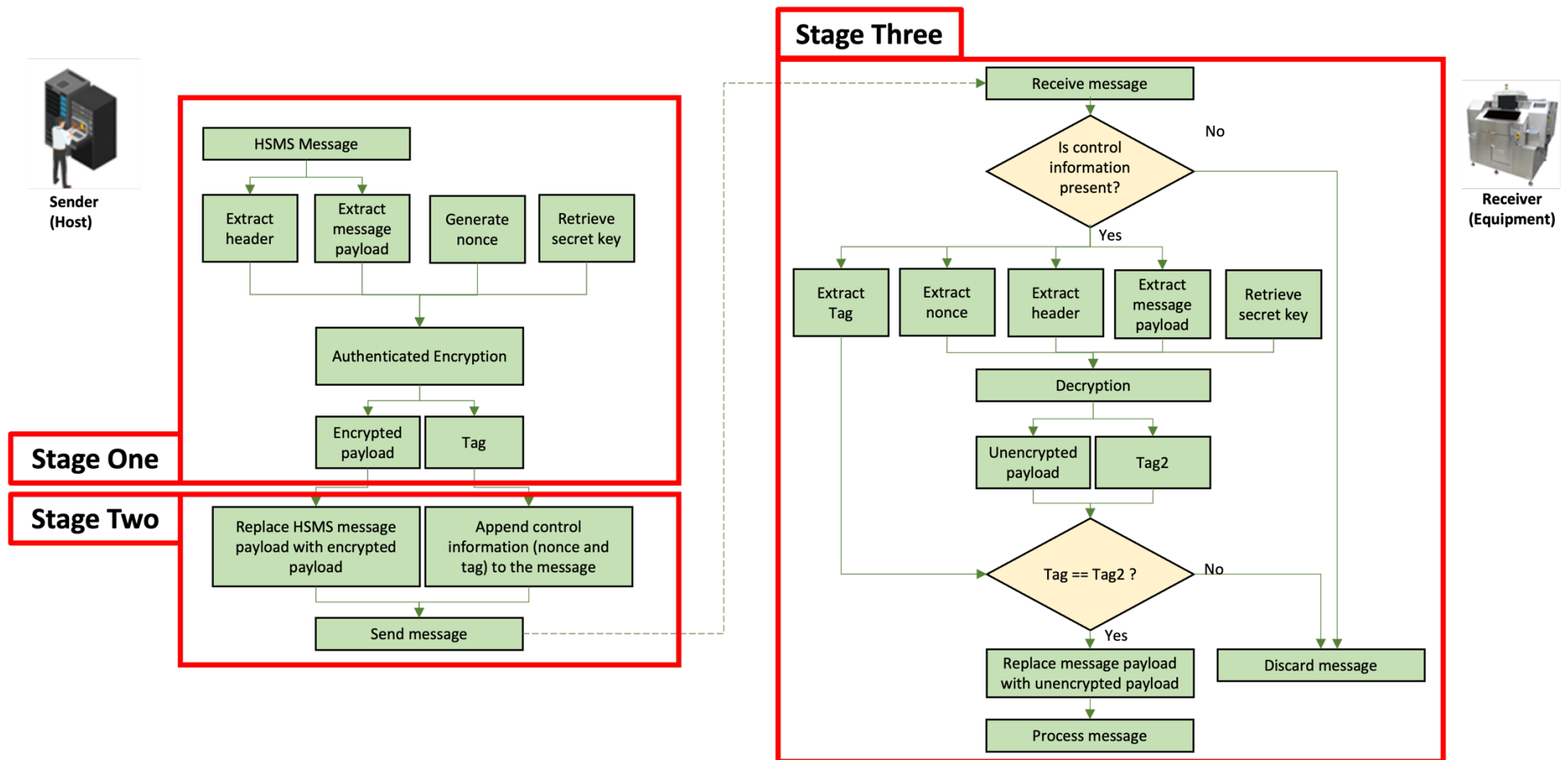


Figure 3.5 Proposed Mechanism Workflow

3.4 Designing Testbed Environment and Assessing Performance

This phase aims to design a testbed environment where the proposed mechanism can be evaluated using different assessment metrics. The experiments aim to determine the processing speed, efficiency, and functioning of the proposed mechanism.

3.4.1 Testbed Design

The proposed mechanism aims to enhance the security of SECS/GEM communication in industrial networks, which often have critical and sensitive equipment vulnerable to cyber-attacks. In this testbed design, the network topology consists of a host, equipment, and a switch. The testbed is designed similar to the related work (S. U. A. Laghari, Manickam, Al-Ani, et al., 2021a). The host and equipment are equipped with the proposed mechanism and set up to log information to evaluate the processing time. The network traffic is captured using Wireshark to dissect and analyze the contents of the messages from an attacker's perspective. The goal is to collect data, analyze and compare the results obtained from the proposed mechanism with other existing mechanisms to determine its effectiveness in enhancing the security of SECS/GEM communications in industrial networks.

3.4.2 Evaluation Matrices

Various assessment metrics are utilized to assess if the proposed SECS/GEM-AE mechanism meets the requirements of this study and to evaluate the proposed SECS/GEM-AE mechanism. These measurements include processing time and control overhead incurred with each message transmitted, which were also used for evaluating the related work (S. U. A. Laghari, Manickam, Al-Ani, et al., 2021a).

3.4.3 Processing Time

Performance is the most important aspect of SECS/GEM because it provides real-time insights into the equipment for effective monitoring, regulation, and decision-making. The SECS/GEM standard states that reports can be generated at a frequency of 1Hz (i.e., one report cycle per second), and there are implementations in which this frequency is 10Hz (Weber, 2016), resulting in ten reports per second. The ultimate design purpose of the SECS/GEM protocol is performance; hence, any mechanism that brings up security features must also be very efficient to impact performance as little as possible. Assessment of the complexity of a security system can be based on the time required to complete the specified task (Bayılmış et al., 2022). Generally, a more complex mechanism is less efficient and harder to maintain.

In contrast, a less complex mechanism can function more efficiently and takes less processing time, incurring less cost on the resources. This study adds message data confidentiality and authentication to SECS/GEM communications in an industrial network for data and control messages transferred between the host and equipment using a less sophisticated and easily deployable mechanism, one of the study's design criteria. Consequently, it is necessary to evaluate the complexity of the proposed method and the behavior of the processes involved in providing security characteristics.

Even though SECS/GEM messages are classified as control and data messages, the sender and receiver of both messages must perform the same steps. The steps include generating a request message on the sender, verifying the request message on the receiver, generating a response message, and verifying the message on the receiver.

In SECS/GEM-AE messages, the sender must encrypt the message payload and generate a tag for verifying the message at the receiver. The control information with

its required fields (nonce, tag) is then appended to the message. Deducting the ending time (TS_{end}) from the starting time (TS_{start}) of the mentioned processes gives the processing time (PT) at the sender side. The processing time at the sender is calculated using 3-1.

$$PT = TS_{end} - TS_{start} \quad 3-1$$

Applying the equation at the sender, PT is the processing time (in milliseconds) for sending messages, TS_{end} is the ending time of generating the SECS/GEM-AE messages at the sender, and TS_{start} is the start time of generating the SECS/GEM-AE message.

Upon receiving a control or data message, the receiver (i.e., the host or the equipment) should decrypt (in case of data message) and perform verification of the message to prevent the injection of spoofed messages that could interrupt the communication between a host and its equipment. For SECS/GEM-AE, the receiver must confirm the presence of the control information on the message before starting with the decryption and message verification. The processing time PT is measured at the receiver's end by subtracting the ending time (TS_{end}) from the starting time(TS_{start}) for the message decryption and verification. The time required to verify SECS/GEM-AE messages at the receiver is computed using 3-1, where PT is the processing time (in milliseconds) at the receiver that needs to verify the SECS/GEM-AE message. TS_{end} is when the verification is completed, and TS_{start} is when the verification is at the receiver.

3.4.4 Control Information Overhead

SECS/GEM-AE appends control information to all messages transmitted by the sender; hence, it adds overhead to each message exchanged between a sender and receiver. The control overhead also adds complexity (Bayılmış et al., 2022), therefore

it is also measured for evaluating SECS/GEM-AE. The control information has a fixed size of 32 bytes, sufficient for piggybacking necessary control data. The control overhead (CO) incurred by SECS/GEM-AE is computed as follows:

$$CO = \frac{CI_{size}}{Message_{size} + CI_{size}} \quad 3-2$$

Where CO is the message overhead computed for each message transmitted or received by SECS/GEM-AE enabled host or equipment, CI_{size} is the size of the control information, and $Message_{size}$ is the size of a standard SECS/GEM message (i.e., control or data message).

3.5 Chapter Summary

This chapter explains the design of the SECS/GEM-AE mechanism for securing SECS/GEM communications in the ecosystem of Industry 4.0 and serves as the core of this research. The SECS/GEM-AE mechanism intends to satisfy the primary design objectives described in Chapter 1 (i.e., to provide data confidentiality, and authenticate messages). The proposed mechanism introduces a set of control information appended to each message and used by the receiver to authenticate and verify the message.

This chapter also included a comprehensive discussion of the theoretical capabilities of the proposed security mechanism's underlying architecture. The proposed SECS/GEM-AE mechanism enables authenticated entities to communicate over an industrial network. The proposed mechanism ensures that messages are not altered in transit and prevents adversaries from obtaining sensitive information in SECS/GEM communications. Experiments are conducted as part of the evaluation process to demonstrate the proposed mechanism's effectiveness and applicability.

Chapter 4 examines the proposed SECS/GEM-AE mechanism's design and implementation details.

CHAPTER 4

DESIGN AND IMPLEMENTATION OF THE PROPOSED SECS/GEM SECURITY MECHANISM (SECS/GEM-AE)

In this chapter, the steps necessary for the implementation of the SECS/GEM-AE mechanism are outlined. These steps include setting up the testbed environment, installing the necessary tools for monitoring and capturing data, and configuring the devices needed to run the proposed mechanism. Moreover, the topology of the experimental industrial network, including the hardware and software used, is presented. Additionally, a computer is introduced to evaluate the security of the proposed mechanism, which is set up to simulate an unauthorized device capable of monitoring SECS/GEM communications over the network.

The remainder of this chapter is divided into four sections. The first section, Section 4.1, outlines the proposed mechanism's prerequisites and explains the packet-capturing process and the Wireshark tool. In Section 4.2, details of the implementation of the SECS/GEM-AE mechanism are discussed. Section 4.3 describes the experimental scenarios used. Section 4.4 concludes and summarizes the chapter.

4.1 Requirements for SECS/GEM-AE Implementation

To assess the SECS/GEM-AE mechanism, a testbed environment was set up at National Advanced IPv6 Centre (NAv6), Universiti Sains Malaysia (USM). This testbed included a host, equipment, and an unauthorized device that could monitor and capture the messages sent between the host and equipment. The specifics of this environment are discussed in Section 4.1.3.

To evaluate the SECS/GEM-AE mechanism, a Python-based SECS/GEM emulator was used to run the host and equipment in the testbed environment. The

SECS/GEM-AE mechanism was integrated into the emulator to piggyback control information with all SECS/GEM messages. Additionally, SECS/GEMsec was implemented in the emulator so that the performance of SECS/GEM-AE could be compared to that of SECS/GEMsec. Further prerequisites for the SECS/GEM-AE implementation are discussed in subsequent sections.

4.1.1 Programming Language

Python is an extremely powerful programming language well-suited for developing prototypes in network research. It is a high-level language that is easy to read, write and understand. It has an extensive set of libraries and frameworks that make it easy to create network-based applications quickly and efficiently. Python also has numerous modules and packages that can be used to create network-based simulations and experiments. Due to its flexibility and ease of use, Python is an ideal language for developing prototypes for network research.

In addition, the PyCharm tool was used to implement SECS/GEM-AE over a Python-based SECS/GEM emulator. PyCharm is an integrated development environment (IDE) for the Python programming language. It is designed to make programming easier and more efficient by providing an intuitive graphical user interface and features like code completion, refactoring, and debugging. PyCharm also supports various plugins for additional features like databases and web frameworks. PyCharm is a powerful and versatile tool many developers use to create, edit, and debug Python code. Figure 4.1 depicts the PyCharm IDE.

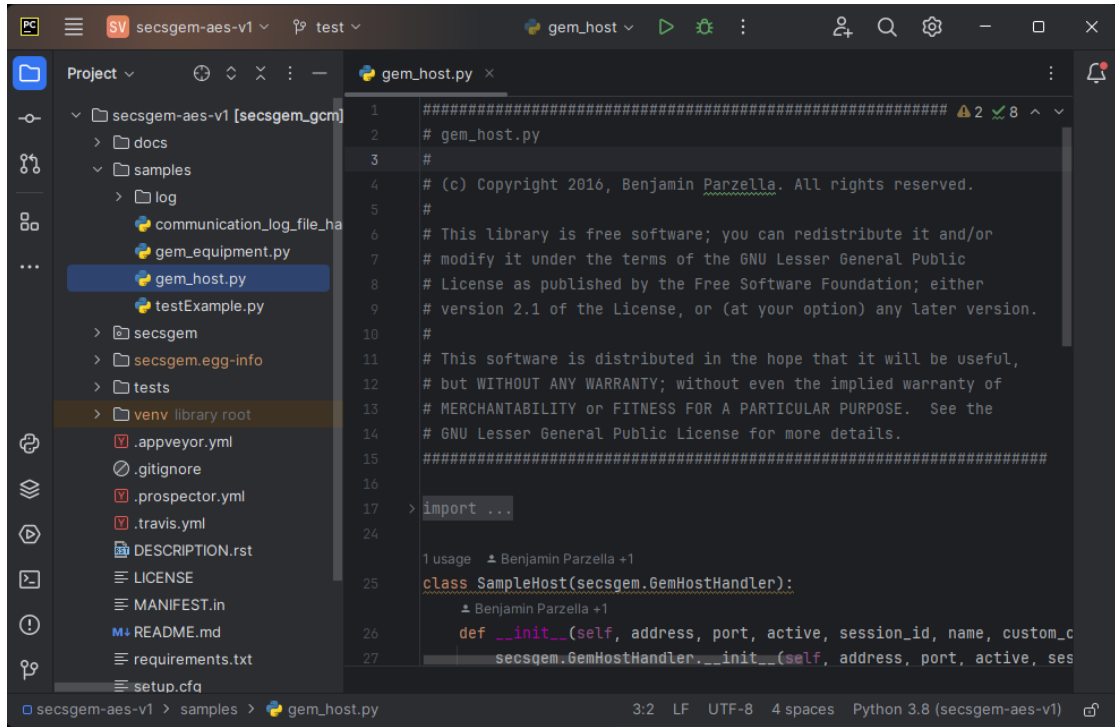


Figure 4.1 PyCharm IDE

The proposed SECS/HEM-AE mechanism is entirely written in Python, including the control information structure and functionality. The added functions are integrated into the existing Python-based SECS/GEM emulator (S. U. A. Laghari, Manickam, Al-Ani, et al., 2021a; Parzella, 2020). An object-oriented approach was adopted, and all the required classes, message translators, and appending procedures were written in Python. By using the same programming language to implement all of the mechanisms, including SECS/GEMsec, an unbiased performance comparison of the proposed SECS/GEM-AE mechanism could be made.

4.1.2 Packet Capturing

Packet analyzers, or packet sniffers, capture and examine packets as they travel through a network device. Therefore, a packet capture tool is necessary to analyze SECS/GEM-AE messages exchanged between hosts and equipment in an industrial network. To this end, Wireshark captures and visualizes the packets exchanged between

the host and the equipment (Chappell & Combs, 2013; Menezes & Stebila, 2021). Further details of Wireshark are discussed in the following subsections.

Wireshark is a powerful network analyzer used for monitoring and analyzing network traffic. It is a cross-platform tool that captures, dissects, and displays packets transmitted over networks. Wireshark supports various protocols, allowing users to filter and search for packets based on various criteria. It also includes various powerful features, such as the ability to decode and reassemble packets, decode protocol fields, and generate protocol statistics. Wireshark is a powerful and versatile tool many network administrators use for various tasks. Wireshark is used to observe the SECS/GEM-AE messages exchanged between the sender and receiver in the testbed environment to investigate the message components (e.g., control information and encrypted payload). Figure 4.2 shows a SECS/GEM message captured and parsed by Wireshark in the testbed environment.

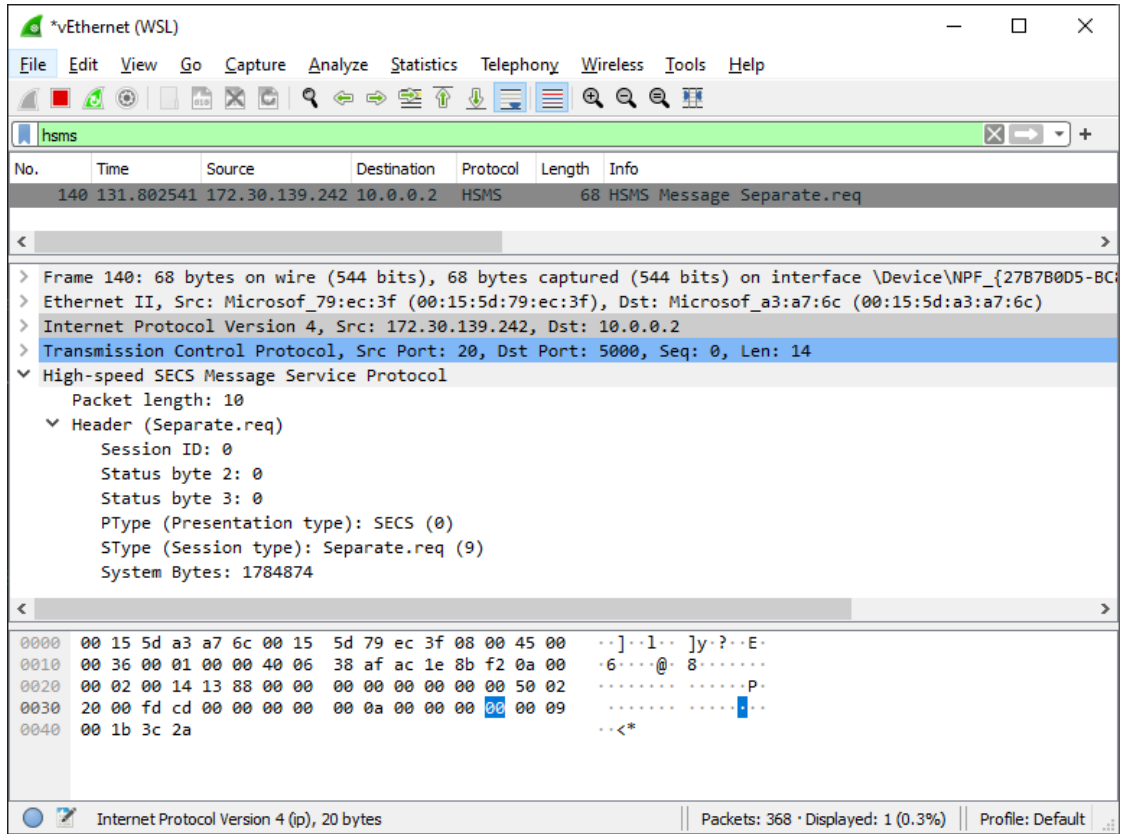


Figure 4.2 SECS/GEM Message Captured on Wireshark

4.1.3 Testbed Environment Setup

The setup of the testbed environment is one of the prerequisites for testing and evaluating the proposed SECS/GEM-AE mechanism. The testbed environment consists of a host, equipment, an unauthorized device, and a network switch. The host and equipment are configured on Ubuntu Linux-based computers, while the unauthorized device is configured on a specialized Linux operating system called Kali Linux. The proposed SECS/GEM-AE is developed following SEMI's SECS/GEM specifications. The Python-based SECS/GEM implementations (Parzella, 2020) are installed on two computers that act as the host and equipment. In the experiment scenario, the unauthorized device is assumed to have bypassed the network security and penetrated the firewall, allowing them to eavesdrop on the communications and sniff packets exchanged between the host and equipment. Figure 4.3 shows a typical industrial

network topology with an unauthorized device present. Table 4.1 shows the hardware and software specifications of the machines in the testbed.

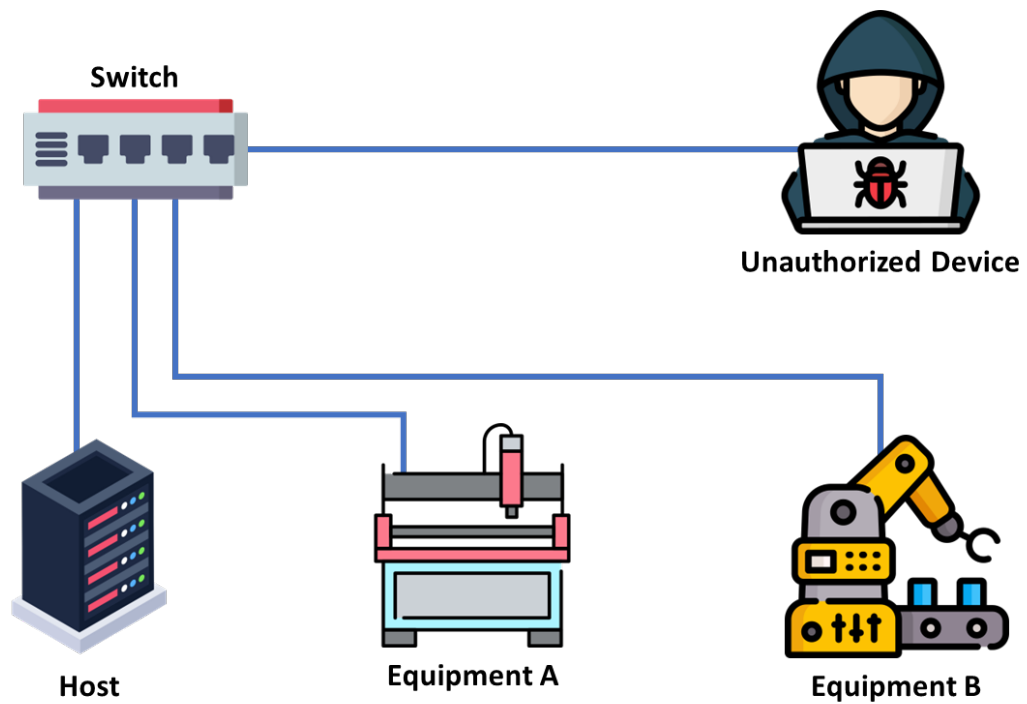


Figure 4.3 Industrial Network Topology with an Unauthorized Device

Table 4.1 Testbed Hardware and Software Specifications

Device Role	Specifications	Memory	Operating System
Host	Intel(R) Core(TM) i7-9750H @ 2.60GHz x 6	8 GB	Ubuntu-18.04
Equipment	Intel(R) Core(TM) i5-6500 @ 3.20 GHz x 4	2 GB	Ubuntu-18.04
Unauthorized device	Intel(R) Core(TM) Ci3-330M @ 2.13GHz x 2	8 GB	Kali Linux 2020.3
Switch	Cisco Catalyst 2960 Fast Ethernet	—	—

4.2 SECS/GEM-AE Implementation Details

The proposed SECS/GEM-AE mechanism necessitates that the sender attaches control information to each message. The receiver extracts the control information and must analyze the contents to decide whether to accept or reject the message. The actions carried out by the sender utilizing SECS/GEM-AE are encrypting the message payload and using SECS/GEM header as authenticated data input, generating the Tag, and attaching the control information (nonce and tag) to the message. The receiver performs the inverse of these steps and decides the message's validity. The subsequent sections explain these processes in detail. Figure 4.4 illustrates a high-level view of the functions executed in the testbed environment by the SECS/GEM-AE-enabled sender and receiver.

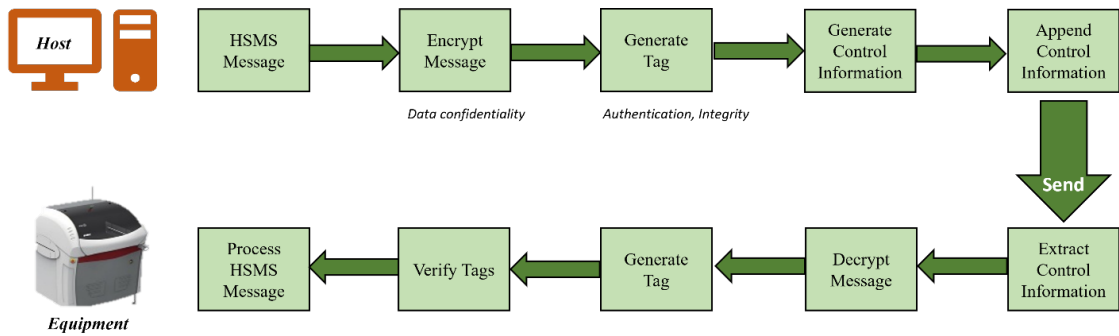


Figure 4.4 High-Level Overview of SECS/GEM-AE Flow

4.2.1 Encrypting SECS/GEM data and Generating Authentication Tag using Authenticated Encryption Algorithms

The first stage is encryption. The secret key is retrieved based on the configuration. A random 16-byte nonce is generated to be used with AES-GCM. AES-GCM takes four inputs: the secret key, nonce, plaintext, and associated data (data that is not encrypted but is authenticated). The plaintext, in this case, is the SECS/GEM payload. The associated data is the SECS/GEM header. The output is the ciphertext and tag. The ciphertext is the encrypted form of the SECS/GEM payload. The tag is a digest generated with the plaintext and associated data used at the receiver to verify the

message's integrity during decryption. After completing the encryption process, the plaintext payload of the SECS/GEM message is replaced with the ciphertext.

4.2.2 Control Information Generation (at the Sender)

The control information is the foundation of the proposed SECS/GEM-AE mechanism. It contains the necessary information to decrypt and verify SECS/GEM-AE messages (Figure 3.4). To generate the control information, the sender has to place the nonce (generated randomly to be used as input for the encryption process) and the tag (generated as an output of the encryption process) as presented in the control information structure. The receiver requires the nonce as the encrypted message payload can only be successfully decrypted and verified if the same nonce is used. The tag is used to verify and authenticate the message by the receiver. The tag is a digest of the encrypted data and the associated data. If the data is changed in any way, the tag resulting from the decryption would be different. Once the control information is generated, it is attached to the end of the message and sent.

4.3 Processing SECS/GEM-AE Messages (at the Receiver)

The machine set up in passive mode opens a port to await incoming requests pertaining to SECS/GEM communications. It will only process those messages that include the SECS/GEM-AE control information and discard all other messages. The operations SECS/GEM-AE performs at the receiver are explained in the following subsections.

4.3.1 Control Information Verification

The receiver must assess the message's authenticity and check it for several conditions. When a message is received on the port address usually set to 5000, it is necessary to determine whether it contains control information. If the message does not

contain control information, it is likely from a device with an incompatible or standard SECS/GEM version, and the message is discarded. It is necessary to perform several checks when control information is appended to the end of a message. The receiver begins by examining whether a nonce is present, which is achieved by analyzing 16 bytes of data following the message. If the nonce is absent, the message is rejected. Subsequently, the receiver examines whether a tag is present by scrutinizing another 16 bytes of data following the nonce. If the tag is not present, the message is discarded. The message is considered valid and forwarded for further processing when all these checks are passed. The pseudocode and flowchart for control information verification at the receiver are provided in Figure 4.5 and Figure 4.6, respectively.

SIB Verification (SECS/GEM-AE Message)	
Receive SECS/GEM-AE request message	
<i>if</i> message does not contain control information, <i>then</i>	
<i>return</i> false	
<i>if</i> Nonce is not present, <i>then</i>	
<i>return</i> false	// nonce not found.
<i>if</i> Tag is not present, <i>then</i>	
<i>return</i> false	// tag not found.
<i>return</i> true	

Figure 4.5 Pseudocode for Control Information Verification at the Receiver

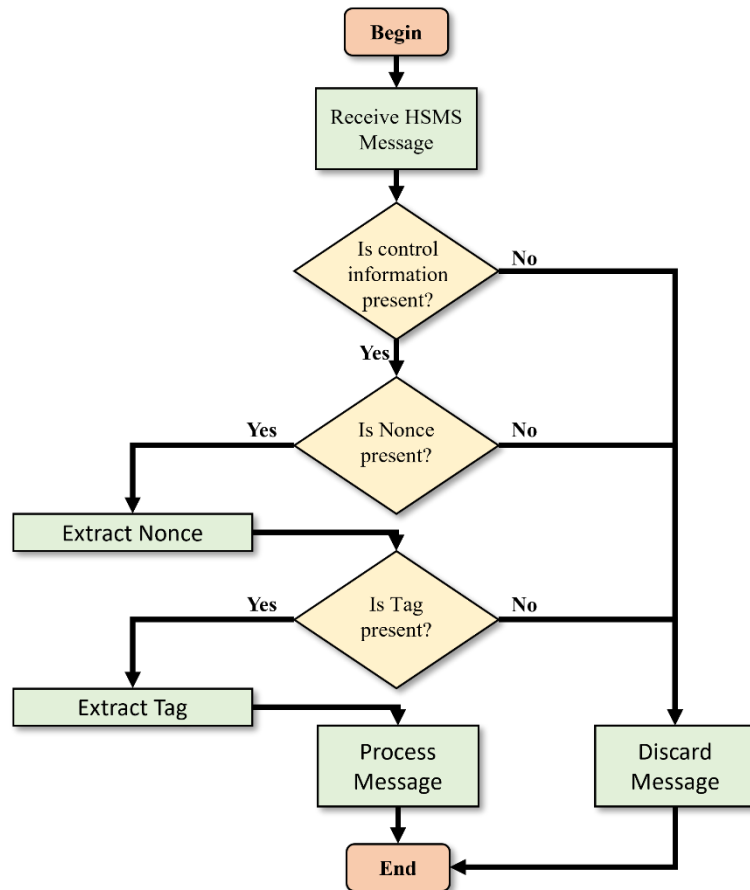


Figure 4.6 Flow of Control Information Verification at the Receiver

4.3.2 Decrypting SECS/GEM Payload and Message Authentication at the Receiver

The SECS/GEM-AE mechanism protects communications over SECS/GEM from unauthorized devices, protecting confidential data and authenticating and verifying messages. The receiver uses the Nonce and Tag (Tag1) from the control information. The decryption requires four inputs: the Secret Key, Nonce, Ciphertext (encrypted payload), and Associated Data (the SECS/GEM Header). The nonce used as input is the same in the control information. The result is two outputs, Plaintext (the original decrypted form of the SECS/GEM payload from the sender) and a second Tag (Tag2). The message is accepted if both Tags (Tag1 and Tag2) match; otherwise, it is discarded. Figure 4.7 and Figure 4.8 display the pseudocode and flowchart for verifying the Tag at the receiver, respectively.

Tag Verification (SECS/GEM-AE Message)

Receive HSMS request message

Tag1 = Extract and save Tag1 from the control information

Tag2, Plaintext = Decrypt Message (Secret-Key, Nonce, Ciphertext,
Associated Data (HSMS Header))

if Tag1 is not equal to Tag2, *then*
 return false

return true

Figure 4.7 Pseudocode for Tag Verification at Receiver

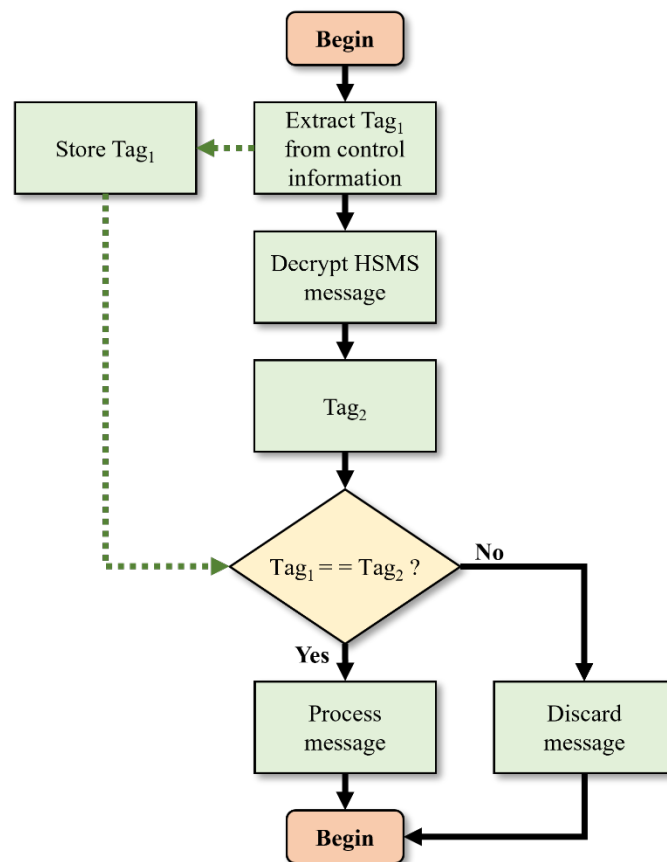


Figure 4.8 Flow of Tag Verification at the Receiver

4.4 Chapter Summary

In this chapter, the implementation of the SECS/GEM-AE mechanism was discussed. To implement the SECS/GEM-AE mechanism, programming language, network traffic capture, and monitoring tools were needed. A python-based

implementation of SECS/GEM was utilized for this study. The testbed environment's network topology, hardware, and software specifications were also discussed. The discussion detailed how SECS/GEM host and equipment were configured and how an attacker machine would be placed to monitor the traffic using the tools introduced in this Chapter. The implementation details of the SECS/GEM-AE mechanism, such as message encryption, verification tag generation, control information generation, control information verification at the receiver, message decryption, and authentication, were all described. The encryption would ensure data confidentiality while the tag would ensure messages are authenticated. The primary intent of the implementation is to show that it is suitable for secure communication in the manufacturing environment without compromising performance. The experiments and the analysis of the results are detailed in Chapter 5.

CHAPTER 5

RESULTS AND DISCUSSION

This chapter discusses the findings obtained while evaluating the SECS/GEM-AE mechanism under the experiments performed. During the experiments, processing time and control overhead are measured for the proposed SECS/GEM-AE mechanism. The experiments are performed 20 times to satisfy the computer science test requirements (Devore, 2015). The results are then analyzed to prove that the proposed SECS/GEM-AE mechanism can be effectively implemented in industrial networks to secure communication over the SECS/GEM communication protocol.

The primary design objective of the SECS/GEM-AE mechanism is to secure SECS/GEM communications by adding data confidentiality and authentication while maintaining performance close to standard SECS/GEM. The results obtained from the experiments are then compared with SECS/GEMsec, and the conclusion is drawn.

The SECS/GEMsec security mechanism as mentioned in Chapter 2 was developed to ensure message authentication and protect communication between equipment and hosts in the semiconductor and other manufacturing industries. SECS/GEMsec addresses the critical vulnerabilities in SECS/GEM by providing Authentication and Message integrity. However, it does not provide data confidentiality and requires high processing time.

The strategy to evaluate the performance of the SECS/GEM-AE mechanism is carried out in two steps:

- (1) Since SECS/GEM-AE supports several authenticated encryption algorithms with varying key sizes, each variant's processing time and control overhead is calculated separately.
- (2) The performance of SECS/GEM-AE is then compared with standard SECS/GEM and other available security mechanisms.

The strategy used to evaluate the proposed mechanism's performance is depicted in Figure 5.1. The rest of the chapter is organized as follows. Section 5.1 discusses the experiments conducted to measure and compare results with other mechanisms regarding processing time and control overhead. Section 5.2 discusses the attainment of objectives in this study, and Section 5.3 summarizes the entire chapter.

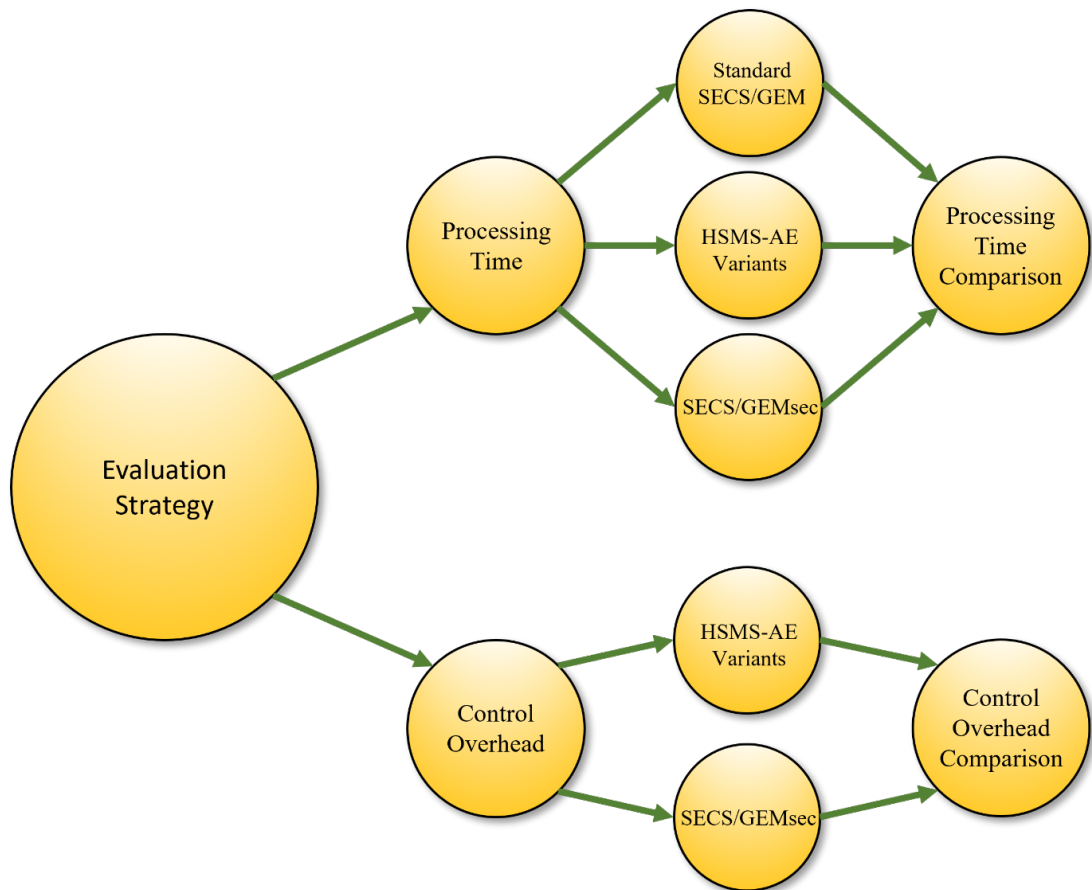


Figure 5.1 Evaluation Strategy

5.1 Performance and Complexity Evaluation Experiments

The experiments evaluate SECS/GEM-AE's processing time and control overhead. Industrial equipment running with SECS/GEM-AE must only communicate with authorized devices; hence, each message must include control information to be validated at the receiver; otherwise, the message is rejected. SECS/GEM-AE takes longer to process than standard SECS/GEM since it performs additional tasks (i.e., encrypting the message data, generating an authentication tag, generating control information, and appending it to the SECS/GEM message). This processing time must be ideal, or SECS/GEM-AE won't be commercially viable. The experiments were repeated 20 times for this evaluation. Processing time is the most important parameter in evaluating SECS/GEM-AE; hence much of this chapter is dedicated to it.

5.1.1 Processing Time

This section describes SECS/GEM-AE's communication stages and the time consumed to process these stages. SECS/GEM-AE generates the control information at the sender, which is used at the receiver end to verify and authenticate the message. In the experiments, the host was configured in active mode and became the sender, while the equipment configured in passive mode became the receiver. There are four stages in SECS/GEM-AE communication, namely generating the message at the sender (Host-S), processing the received message at the receiver (Equip-R), generating the appropriate reply message at the receiver (Equip-S), processing the reply message at the host (Host-R). Table 5.1 describes the processes involved in the sender's and receiver's side during the four stages.

Table 5.1 Processes Involved at the Sender and Receiver

Stage	Processes
Host-S, Equip-S	1. Encrypt SECS/GEM message payload. 2. Generate authentication tag. 3. Generate control information and append it to the SECS/GEM message.
Equip-R, Host-R	1. Extract control information. 2. Decrypt SECS/GEM message payload. 3. Verify and authenticate messages using the authentication tag.

The aforementioned processes add overhead to SECS/GEM-AE, resulting in increased processing time. The design consideration of standard SECS/GEM protocol is performance. Hence the proposed mechanism must not incur heavy computational or processing overhead which would ultimately make it less likely useful for the production setups. Thus, the processing time must be evaluated to determine the efficiency of SECS/GEM-AE.

The processing time for the mentioned processes at each stage is computed by first storing the current timestamp of the system in a variable T_{start} . Once the processes are completed, the current timestamp is saved in a variable T_{end} . The processing time PT is computed by subtracting the starting time T_{start} from the ending time T_{end} .

SECS/GEM-AE can support a variety of authenticated encryption algorithms and key sizes. Only AES-GCM with different key sizes has been used in this study, namely AES-GCM-128, and AES-GCM-256. The processing time of SECS/GEM-AE is compared with standard SECS/GEM and the compared work (SECS/GEMsec). The mechanisms that are closer to the processing time of the standard SECS/GEM processing time are considered the best possible security mechanism. The following sub-sections evaluate, compare, and present the results of SECS/GEM-AE.

5.1.1(a) Sending Messages (Host-S)

This section presents the processing time for SECS/GEM-AE with AES-GCM-128 and AES-GCM-256 to send a message from the host. Appendix A presents the processing time taken by each SECS/GEM-AE variant to encrypt the SECS/GEM payload, generate the authentication tag, generate control information, and append it to the SECS/GEM message.

Table 5.2 shows the minimum and maximum processing time, mean and standard deviation of the processing time of SECS/GEM-AE from the experiments. It is seen that SECS/GEM-AE with the AES-GCM-128 algorithm consumes less time to process than with the AES-GCM-256 algorithm. AES-GCM-128 achieves an average processing time of approximately 1.12 milliseconds due to its smaller key size (128 bits). AES-GCM-256, on the other hand, achieves an average processing time of approximately 1.25 milliseconds, which is expected as it uses a larger key size (256 bits) and involves more rounds of processing. The processing time stayed well within 1.50 milliseconds in both cases, which is exceptional for a mechanism that provides message data confidentiality and authentication.

Table 5.2 SECS/GEM-AE Host-S Processing Time Analysis

	SECS/GEM-AE (AES-GCM-128)	SECS/GEM-AE (AES-GCM-256)
Min	1.091	1.238
Max	1.150	1.278
Mean	1.123	1.255
SD	0.019	0.013

Figure 5.2 visualizes the processing time of SECS/GEM-AE with AES-GCM-128 and AES-GCM-256. The performance difference can be seen clearly in the figure,

with AES-GCM-256 having a higher processing time. However, AES-GCM-256 also uses a bigger key, double the key size used by AES-GCM-128. The bigger key size makes AES-GCM-256 more secure and robust than the AES-GCM-128, with a slight 0.13-millisecond increase in the average processing time. SECS/GEM-AE with AES-GCM-128 can be considered in scenarios where performance is the most important factor for the system.

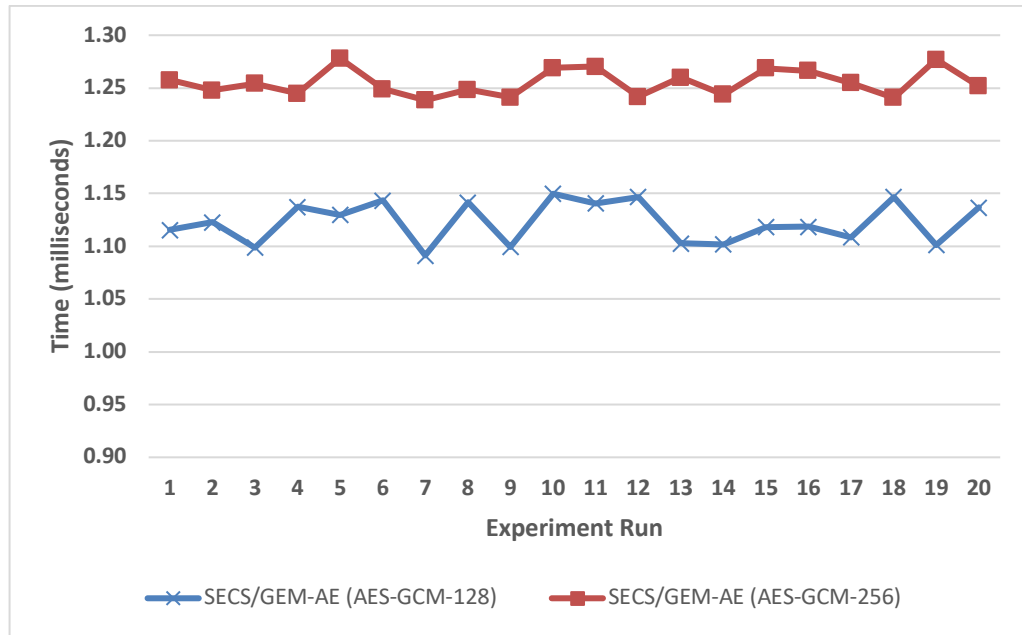


Figure 5.2 SECS/GEM-AE AES-GCM-128 vs AES-GCM-256 (Host-S)

5.1.1(b) Receiving Message (Equip-R)

This section presents the processing time for SECS/GEM-AE with AES-GCM-128 and AES-GCM-256 to process a message received at the equipment. Appendix B shows the processing time by each SECS/GEM-AE variant to extract control information, decrypt the encrypted SECS/GEM payload, and verify and authenticate the message using the authentication tag.

Table 5.3 shows the minimum and maximum processing time, mean and standard deviation of the processing time of SECS/GEM-AE from the experiments. It

is seen that SECS/GEM-AE with the AES-GCM-128 algorithm consumes less time to process than with the AES-GCM-256 algorithm. AES-GCM-128 achieves an average processing time of approximately 0.63 milliseconds due to its smaller key size (128 bits). AES-GCM-256, on the other hand, achieves an average processing time of approximately 0.73 milliseconds, which is expected as it uses a larger key size (256 bits) which involves more rounds of processing. The processing time stayed well within 1 millisecond in both cases, which is exceptional for a mechanism that provides message data confidentiality and authentication.

Table 5.3 SECS/GEM-AE Equip-R Processing Time Analysis

	SECS/GEM-AE (AES-GCM-128)	SECS/GEM-AE (AES-GCM-256)
Min	0.617	0.696
Max	0.650	0.786
Mean	0.634	0.738
SD	0.010	0.030

Figure 5.3 visualizes the processing time of SECS/GEM-AE with AES-GCM-128 and AES-GCM-256. The performance difference can be seen clearly in the figure, with AES-GCM-256 having a higher processing time. However, AES-GCM-256 also uses a bigger key, double the key size used by AES-GCM-128. The bigger key size makes AES-GCM-256 more secure and robust than the AES-GCM-128, with a slight 0.1-millisecond increase in the average processing time. SECS/GEM-AE with AES-GCM-128 can be considered in scenarios where performance is the most important factor for the system.

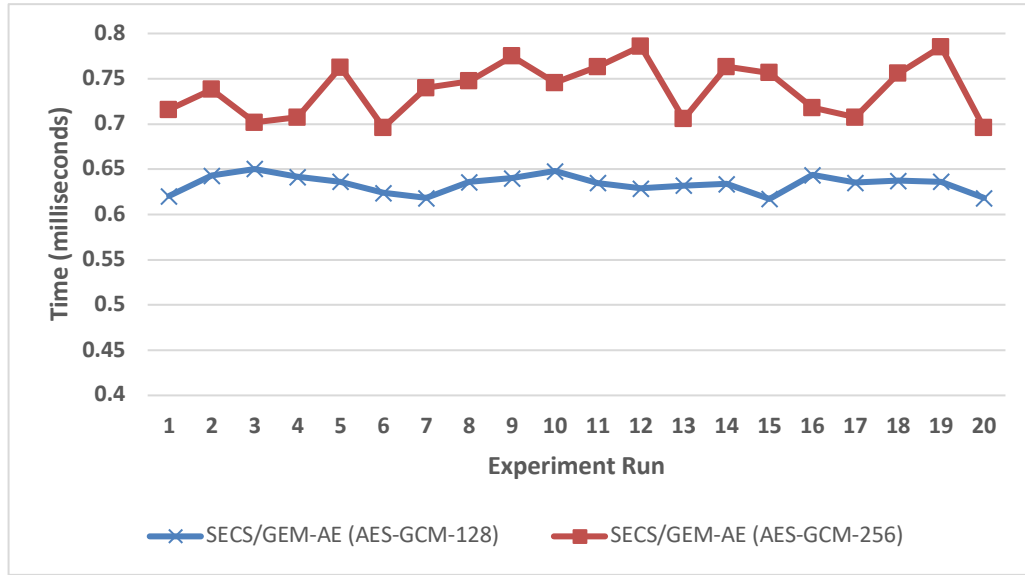


Figure 5.3 SECS/GEM-AE AES-GCM-128 vs AES-GCM-256 (Equip-R)

5.1.2 Processing Time of SECS/GEM-AE vs. SECS/GEMsec

In this section, SECS/GEM-AE is compared with SECS/GEMsec (S. U. A. Laghari, Manickam, Al-Ani, et al., 2021b) in terms of processing time. The processing time is measured using the formula 3-1 discussed in Section 3.4.3. SECS/GEM-AE is designed with message data confidentiality and authentication in mind. SECS/GEMsec is also designed similarly; however, its focus is limited to message authentication and integrity. It also protects SECS/GEM communications against cyber-attacks utilizing modified messages. However, SECS/GEMsec does not provide data confidentiality for SECS/GEM communications. The following sub-sections compare the processing time of SECS/GEM-AE with SECS/GEMsec when sending and receiving messages.

5.1.2(a) Sending Messages

To achieve data confidentiality and authenticate messages, SECS/GEM-AE uses the control information to transfer additional information with every message. As mentioned in the previous sections, sending a message involves encrypting the message payload, generating the authentication tag, generating the control

information, and appending it to the message. Similarly, SECS/GEMsec appends a signature to messages, which is used at the receiver's end to authenticate the message. As such, these additional processes incur computational overhead and require additional processing time compared to standard SECS/GEM.

Table 5.4 presents the minimum, maximum, and average processing time of sending SECS/GEM-AE messages using AES-GCM-128 against SECS/GEMsec and standard SECS/GEM. It is seen that standard SECS/GEM's average processing time is 0.67 milliseconds. Standard SECS/GEM's processing time is taken as a benchmark to evaluate the performance of SECS/GEM-AE and SECS/GEMsec. SECS/GEM-AE's average processing time is around 1.12 milliseconds, while SECS/GEMsec's average processing time is around 19.25 milliseconds. SECS/GEM-AE is faster than SECS/GEMsec by approximately 18.13 milliseconds which is quite a large number in terms of processing time. SECS/GEM-AE also achieves a processing time much closer to standard SECS/GEM with a minor increase of 0.45 milliseconds which makes SECS/GEM-AE a much more suitable candidate as a security mechanism for SECS/GEM communications. Figure 5.4 visualizes the processing time of SECS/GEM-AE (AES-GCM-128) against standard SECS/GEM and SECS/GEMsec.

Table 5.4 Processing Time Analysis (Host-S): Standard SECS/GEM vs. SECS/GEM-AE (AES-GCM-128) vs. SECS/GEMsec

	SECS/GEM-AE (AES-GCM-128)	SECS/GEMsec	Standard SECS/GEM
Min	1.091	18.956	0.654
Max	1.150	19.990	0.693
Average	1.123	19.254	0.673

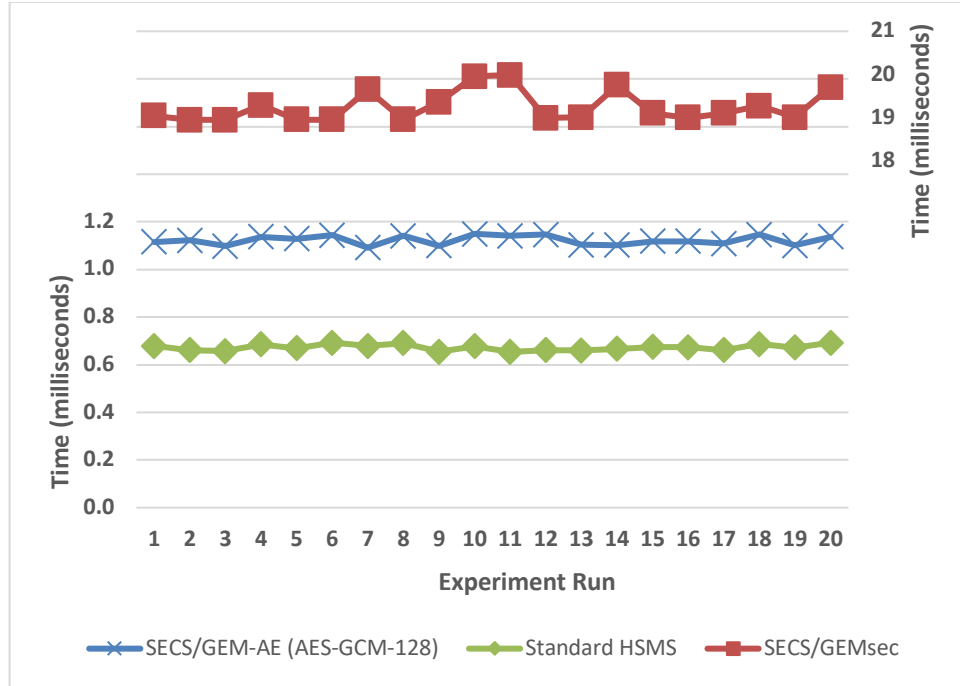


Figure 5.4 Processing Time (Host-S): Standard SECS/GEM vs. SECS/GEM-AE (AES-GCM-128) vs. SECS/GEMsec

5.1.2(b) Receiving Messages

As mentioned in the previous sections, in SECS/GEM-AE-enabled communications, receiving a message involves extracting the control information, decrypting the message payload, and authenticating the message using the authentication tag. SECS/GEMsec similarly extracts the signature attached to the messages to authenticate the message at the receiver. These additional processes require further processing time compared to standard SECS/GEM. This section compares the processing time of SECS/GEM-AE against standard SECS/GEM and SECS/GEMsec.

Table 5.5 presents the minimum, maximum, and average processing time of receiving a SECS/GEM-AE message using AES-GCM-128 against SECS/GEMsec and standard SECS/GEM. It is seen that standard SECS/GEM’s average processing time is 0.046 milliseconds. Standard SECS/GEM’s processing time is taken as a

benchmark to evaluate the performance of SECS/GEM-AE and SECS/GEMsec. SECS/GEM-AE's average processing time is around 0.63 milliseconds, while SECS/GEMsec's average processing time is around 20.38 milliseconds. SECS/GEM-AE is faster than SECS/GEMsec by approximately 19.75 milliseconds which is quite a large number in terms of processing time. SECS/GEM-AE also achieves a processing time much closer to standard SECS/GEM with a minor increase of 0.58 milliseconds which makes SECS/GEM-AE a much more suitable candidate as a security mechanism for SECS/GEM communications. Figure 5.5 visualizes the processing time of SECS/GEM-AE (AES-GCM-128) against standard SECS/GEM and SECS/GEMsec.

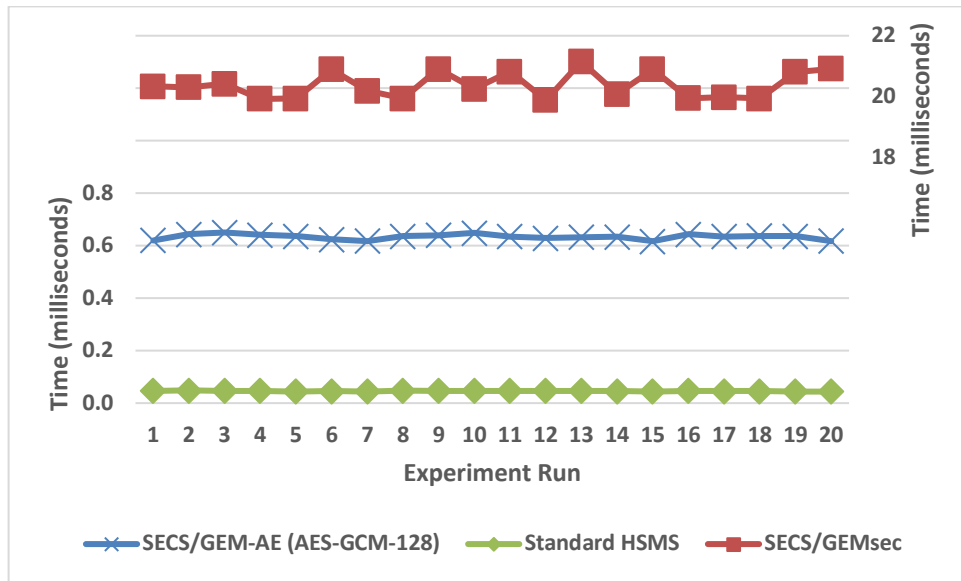


Figure 5.5 Processing Time (Equip-R): Standard SECS/GEM vs. SECS/GEM-AE (AES-GCM-128) vs. SECS/GEMsec

Table 5.5 Processing Time Analysis (Equip-R): Standard SECS/GEM vs. SECS/GEM-AE(AES-GCM-128) vs. SECS/GEMsec

	SECS/GEM-AE (AES-GCM-128)	SECS/GEMsec	Standard SECS/GEM
Min	0.617	19.894	0.044
Max	0.650	21.154	0.048
Average	0.634	20.386	0.046

5.1.2(c) Processing Time Summary

Each of the 20 experiments yielded a slightly different processing time. Table 5.6 presents the average processing time (mean), standard deviation (SD), and overhead for encrypting data, authenticating, and verifying messages. Overhead is the processing time induced by a security mechanism with respect to the mean processing time of standard SECS/GEM.

The Min and Max values show the best and worst processing times for Host-S and Equip-R for various mechanisms. SD shows how far the processing times diverge from the mean value. The SD value represents experiment consistency. As SD approaches zero, outcomes become more consistent. The further the value from zero, the outcomes become less consistent. Table 5.6 shows that SD is negligible for all methods except SECS/GEMsec, which takes 0.35 and 0.43 milliseconds to send and receive messages. Although insignificant, these values are greater than other approaches, possibly due to the complicated RSA mechanism utilized to generate signatures. SD measurements for all mechanisms in this study have shown remarkable consistency in sending and receiving messages (i.e., data and control messages).

$$PT_{overhead} = Mean(PT_{mechanism}) - Mean(PT_{standardHSMS}) \quad (5-1)$$

Table 5.6 Processing Time Overhead

		SECS/GEM-AE (AES-GCM-128)	SECS/GEMsec	Standard SECS/GEM
Host-S	Min	1.091	18.956	0.654
	Max	1.150	19.990	0.693
	Mean	1.123	19.254	0.673
	SD	0.019	0.359	0.013
	Overhead	0.450	18.581	N/A
Equip-R	Min	0.617	19.894	0.044
	Max	0.650	21.154	0.048
	Mean	0.634	20.386	0.046
	SD	0.010	0.432	0.001
	Overhead	0.587	20.339	N/A

The processing time overhead is calculated using (5-1) where PT_{overhead} is the processing time overhead, $Mean(PT_{\text{mechanism}})$ is the mean processing time of the mechanism, and $Mean(PT_{\text{standardSECS/GEM}})$ is the mean processing time of standard SECS/GEM. SECS/GEM-AE has the lowest processing time overhead, 0.45 milliseconds for a host's request message and 0.58 milliseconds for equipment to process the received message. SECS/GEMsec performs worst because it has 18.58 and 20.33 milliseconds overhead to send and receive messages. From this processing time analysis, it can be confirmed that SECS/GEM-AE is the best candidate for secure SECS/GEM communications against SECS/GEMsec in terms of processing time. Thus, the third objective of this study is achieved as SECS/GEM-AE is efficient in terms of processing time compared to the related work and achieving performance close to that of standard SECS/GEM.

5.1.3 Control Overhead

The standard SECS/GEM protocol is highly efficient, considering it uses binary encoding and has data structures such as integers, long integers, floating point values, ASCII text, lists, and so on. These data structures pack data appropriately and efficiently, minimizing bandwidth consumption. As such, it is necessary that the security mechanisms proposed for standard SECS/GEM cause as little overhead as possible. However, SECS/GEM-AE and SECS/GEMsec incur overhead in the form of the control information passed along with messages. The control overhead is measured using the formula 3-2 discussed in Section 3.4.4. The following sub-sections evaluate the control overhead of SECS/GEM-AE and SECS/GEMsec.

5.1.3(a) SECS/GEMsec Control Overhead

SECS/GEMsec sends a signature to authenticate the sender and prevent message alteration. Additional bytes (signature) added to each message represent an overhead compared to conventional SECS/GEM. This increase in message size is called control overhead. Equation (5-2) quantifies this control overhead for SECS/GEMsec (S. U. A. Laghari, Manickam, Al-Ani, et al., 2021a).

$$Overhead\% = \frac{KeySize}{HMS_{hdr} + HMS_{payload} + KeySize} \times 100 \quad (5-2)$$

SECS/GEM_{hdr} is a fixed-sized SECS/GEM header, and SECS/GEM_{payload} is the message's contents. SECS/GEM_{hdr} and SECS/GEM_{payload} are assessed independently since control messages only carry the header. KeySize determines the signature's RSA key size. The signature's size depends on the key size; for example, a 2048-bit key would add 256 bytes of control overhead to every message between the nodes. Control request messages have minimal message sizes, generally just a 14-byte header (i.e.,

four-length bytes and 10 bytes for SECS/GEM header). Every 14-byte header-only message incurs 256 bytes of message signature. The payout is secure communication. The SECS/GEMsec mechanism's message size and control overhead are shown in Table 5.7 and Table 5.8 for key sizes of 2048 and 4096 bits, respectively. As shown in the tables, the overhead decreases as the message size increases, yet it negatively impacts performance when employed with small-sized messages.

Table 5.7 SECS/GEMsec Control Overhead with RSA 2048-bit Key

Message Type	SECS/GEM (header)	Payload (bytes)	Key size (bytes)	Total Size	Control Overhead
Control Message	14	0	256	270	94.815%
Data Message (1KB)	14	1024	256	1294	19.784%
Data Message (1MB)	14	1048576	256	1048576	0.024%
Data Message (10MB)	14	10485760	256	10485760	0.002%

Table 5.8 SECS/GEMsec Control Overhead with RSA 4096-bit Key

Message Type	SECS/GEM (header)	Payload (bytes)	Key size (bytes)	Total Size	Control Overhead
Control Message	14	0	512	526	97.338%
Data Message (1KB)	14	1024	512	1550	33.032%
Data Message (1MB)	14	1048576	512	1049102	0.049%
Data Message (10MB)	14	10485760	512	10486286	0.005%

5.1.3(b) SECS/GEM-AE – Control Overhead

SECS/GEM-AE uses the control information to transmit information to decrypt, authenticate and verify the messages at the receiver. The additional bytes (overhead) in the case of SECS/GEM-AE are the size of the control information. Equation (5-3) quantifies this control overhead for SECS/GEM-AE.

$$Overhead\% = \frac{CISize}{HSMS_{hdr} + HSMS_{payload} + CISize} \times 100 \quad (5-3)$$

The control overhead is calculated using (5-3) where SECS/GEM_{hdr} is a fixed-sized SECS/GEM header, and SECS/GEM_{payload} is the message's contents. SECS/GEM_{hdr} and SECS/GEM_{payload} are assessed independently since control messages only carry the header. The control information contains the nonce (16-Bytes) and authentication tag (16-Bytes), which comprise 32 Bytes. Unlike SECS/GEMsec where the signature's size depends on the key size, SECS/GEM-AE has fixed control information (CISize) in all variants. Control request messages have minimal message sizes, generally just a 14-byte header (i.e., four-length bytes and 10 bytes for

SECS/GEM header). Every 14-byte header-only message incurs 32 bytes of Control information. The payout is secure communication. SECS/GEM-AE's control overhead for various message sizes is shown in Table 5.9. As shown in Table 5.9, the overhead decreases as the message size increases, yet it negatively impacts performance when employed with small-sized messages. However, SECS/GEM-AE achieves relatively less overhead, even for small messages, compared to SECS/GEMsec.

Table 5.9 SECS/GEM-AE Control Overhead

Message Type	SECS/GEM (header)	Payload (bytes)	Overhead (bytes)	Total Size (bytes)	Control Overhead (%)
Control Message	14	0	32	46	69.5652
Data Message(1KB)	14	1024	32	1070	2.9907
Data Message (1MB)	14	1048576	32	1048622	0.0031
Data Message(10MB)	14	10485760	32	10485806	0.0003

5.1.3(c) Control Overhead: SECS/GEM-AE vs. SECS/GEMsec

Standard SECS/GEM has no control overhead because it lacks security features and doesn't provide control data to the receiver for verification. As described in the previous subsections, the SECS/GEM-AE and SECS/GEMsec add control information to the communication, causing control overhead. Table 5.10 presents control overhead across all analyzed mechanisms.

Table 5.10 Control Overhead Comparison

Protocol	Control Message (10 bytes)		Data Message (100 bytes)	
	Actual	Overhead	Actual	Overhead
Standard SECS/GEM	14	0	114	0
SECS/GEMsec	14	512	626	512
SECS/GEM-AE	14	32	146	32

Table 5.10 shows control and data message (10-byte, 100-byte payload) overhead for standard SECS/GEM, SECS/GEMsec, and SECS/GEM-AE. Data messages with 100-byte payload was used so that the results are directly comparable to the related work, SECS/GEMsec (S. U. A. Laghari, Manickam, Al-Ani, et al., 2021a). Large key size gives SECS/GEMsec the highest control overhead (i.e., 4096-bit key). SECS/GEM-AE secures data and control messages from being passively monitored or modified by unauthorized devices with a 32-byte control overhead in all variants. SECS/GEM-AE's control overhead drops rapidly as the message size increases; nonetheless, SECS/GEMsec control overhead with a payload size of 1000 bytes is about 34%, which is high compared to SECS/GEM-AE. Figure 5.6 presents a plot of how message size affects control overhead. Thus, the third objective of this study is achieved as SECS/GEM-AE has less overhead compared to the related work and therefore incurring minimal overhead.

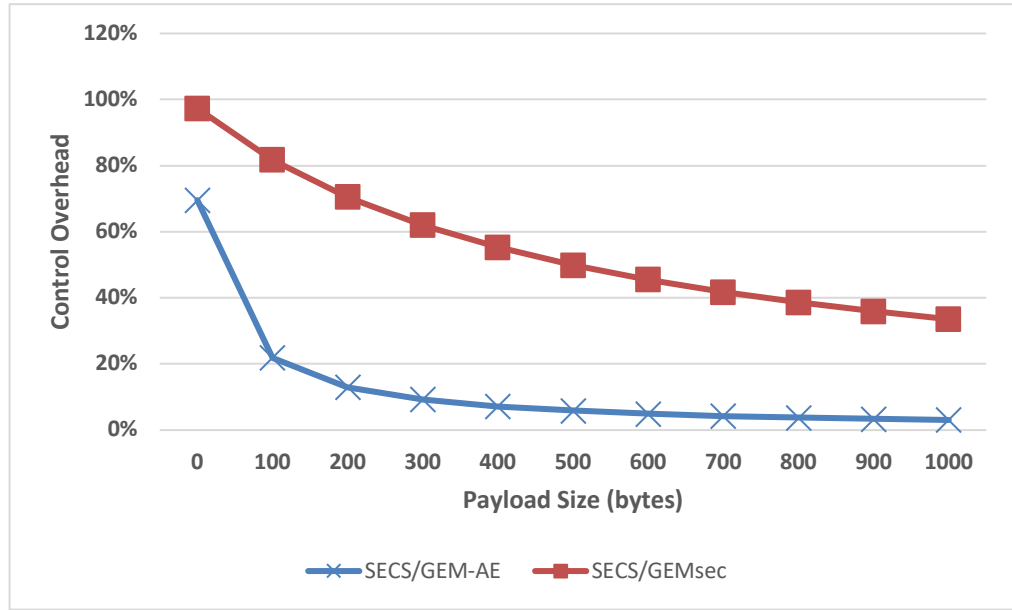


Figure 5.6 Control Overhead vs. Message Size

5.2 Discussion

The previous sections compared SECS/GEM-AE's processing time and control overhead against SECS/GEMsec and standard SECS/GEM. Experiments were run to establish if the proposed SECS/GEM-AE mechanism meets the research objectives established in Chapter 1 (Section 1.5). The following sub-sections evaluate SECS/GEM-AE on these criteria and discuss how message authentication and confidentiality are attained.

5.2.1 Attainment of Authentication

A security mechanism must ensure that only approved SECS/GEM-enabled devices communicate. SECS/GEM-AE appends the control information to each message so the receiver can validate it. Sender uses authenticated encryption methods to encrypt data and generate the authentication tag using the pre-shared secret key. The receiver decrypts the message using the same pre-shared key, generates the tag, and compares it to the tag received in the control information. Messages are processed if the authentication tag matches. Without the key, unauthorized devices cannot

communicate with the receiver; therefore, decrypting a message's data or forging an authentication tag is not possible. Thus, SECS/GEM-AE ensures only messages from authenticated devices are accepted. Thus, the proposed mechanism successfully achieves the first objective of this research.

5.2.2 Attainment of Confidentiality

A security mechanism for SECS/GEM must provide data confidentiality. Since these messages carry sensitive information such as recipes, machine status, and more, it is essential to have data confidentiality. SECS/GEM-AE encrypts the message data using the pre-shared key, changing the plaintext data into ciphertext. This ciphertext is useless to any attacker passively listening on the network as the sensitive information can only be made sense after decryption. The attacker cannot decrypt the ciphertext without the secret key, and breaking the key is computationally infeasible. Therefore, SECS/GEM-AE ensures data confidentiality in SECS/GEM communication messages. Thus, the proposed mechanism successfully achieves the second objective of this research.

5.3 Chapter Summary

This chapter presents the experimental results of SECS/GEM-AE and other security mechanisms, exploring their performance in terms of processing time and control overhead generated. The evaluation involves a comparison of SECS/GEM-AE with standard SECS/GEM and SECS/GEMsec. SECS/GEM-AE demonstrates consistent high performance, exhibiting lower processing time and reduced overhead compared to SECS/GEMsec. In contrast, standard SECS/GEM proves ineffective in preventing cyberattacks within the testbed environment. While SECS/GEMsec successfully provides message authentication, it falls short in providing

confidentiality, and its operational speed is impractical for commercial use. The experiments indicate that SECS/GEM-AE emerges as the most suitable production level security mechanism, offering both data confidentiality and authentication. The concluding chapter delves into the limitations of SECS/GEM-AE and outlines ideas for future research.

CHAPTER 6

CONCLUSION AND FUTURE WORK

This chapter summarizes some of the issues raised and discussed in the current study. This chapter also provides suggestions for future works to enhance the proposed SECS/GEM-AE mechanism. This chapter is organized as follows: Section 6.1 presents the research conclusion, and Section 6.2 discusses possible future work.

6.1 Conclusion

The manufacturing industry is embracing Industry 4.0 and moving towards automation technologies, integrating robotics, machine learning, M2M, and more. M2M communication has helped factory equipment communicate with each other, optimizing and boosting production efficiency. SECS/GEM is at the heart of the semiconductor manufacturing industry. However, SECS/GEM built with only efficiency in mind lacks security for use in an Industry 4.0 environment. It suffers from issues such as the lack of (a) data confidentiality and (b) authentication, which, when left unchecked, can lead to catastrophe in an Industry 4.0 environment. This research addresses these issues and enables SECS/GEM to be used in an Industry 4.0 ecosystem.

SECS/GEMsec, a mechanism proposed to address some of the issues of SECS/GEM was reviewed in detail. SECS/GEMsec used a digital signature-based solution to verify and authenticate the messages on the receiving end. However, it only solves the problem of message authentication in SECS/GEM communications.

A novel security mechanism, SECS/GEM-AE, was proposed and evaluated in this research. SECS/GEM-AE addresses both data confidentiality and authentication

issues in SECS/GEM. Table 6.1 summarizes the mapping of research objectives with the objective attainment.

Table 6.1 Research Objectives and Objective Attainment

Research Objective	Objective Attainment	Section
To provide message authentication and data integrity	The first research objective is achieved by employing Authenticated Encryption	3.2.1 Authenticated Encryption (Stage 1)
To provide data confidentiality	The second research objective is achieved by employing Authenticated Encryption	3.2.1 Authenticated Encryption (Stage 1)
To achieve minimal overhead and high efficiency to provide performance close to the standard SECS/GEM	The proposed mechanism demonstrated performance at par the standard SECS/GEM implementation	5.1 Performance and Complexity Evaluation Experiments

The proposed SECS/GEM-AE mechanism utilizes authenticated encryption to ensure data confidentiality and authentication for messages. Authenticated encryption algorithms encrypt data and generate a tag, which is used to verify the integrity of the data. This process occurs in Stage 1 of the SECS/GEM-AE mechanism, where the message is encrypted, and a tag is generated to authenticate the messages. Consequently, the proposed mechanism successfully accomplishes the first and second objectives of this study. The proposed SECS/GEM-AE mechanism was compared with SECS/GEMsec in terms of processing time and control overhead while using standard SECS/GEM as the benchmark. As seen in Section 5.1, pertaining to performance and complexity evaluation experiments, SECS/GEM-AE surpassed SECS/GEMsec with significantly less processing time and generated less control overhead, thereby achieving the third objective of this study. SECS/GEM-AE's results were much closer to standard SECS/GEM while maintaining good performance and lesser complexity

than SECS/GEMsec. The proposed SECS/GEM-AE mechanism is deemed more efficient and preferable for application in an Industry 4.0 ecosystem based on performance comparisons. Since SECS/GEM-AE's performance is very close to standard SECS/GEM, it can provide real time insights into the manufacturing environment just like standard SECS/GEM.

This study was limited to using AES-GCM authenticated encryption, which is generally fast in hardware implementations. However, certain other algorithms could also be studied for situations where only software implementations of authenticated encryption algorithms are available. This study is also limited to host and equipment that are preconfigured with preferred authenticated encryption and keys and does not consider any negotiation mechanisms or key management mechanisms. These limitations and the methods to overcome them are discussed in the next section.

6.2 Future Work

Although SECS/GEM-AE achieves the objectives of this research work, it can still be further improved. Such improvements can help SECS/GEM communications perform even more efficiently and be applicable for more specific Industry 4.0 scenarios. Following are the suggestion for future works on SECS/GEM-AE:

1. A more flexible SECS/GEM-AE: In this research, SECS/GEM-AE uses AES-GCM with different key sizes. However, SECS/GEM-AE can be more flexible, meaning it could have the ability to choose different encryption algorithms. Another option could be to choose between authentication, encryption, or both depending on the use case.
2. A standardized header for SECS/GEM-AE: The proposed mechanism appends the Nonce and Tag to the end of the message; however, this

can be designed into a standardized header with more options. For example, the options mentioned in the previous point may be included as fields in the SECS/GEM-AE header.

3. Further research on different types of authenticated encryptions, such as stream ciphers.
4. Secret keys are stored in a secure place on the local machine; however, if these keys are compromised, it would result in compromising the entire system. Hence, there is a need for a sophisticated mechanism that makes key management more efficient.

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APPENDICES

APPENDIX A SECS/GEM-AE HOST-S PROCESSING TIME

No.	SECS/GEM-AE (AES-GCM-128)	SECS/GEM-AE (AES-GCM-256)
1	1.115447363	1.257565801
2	1.122689014	1.247635668
3	1.098696716	1.254208177
4	1.137426695	1.244719742
5	1.129690106	1.278101306
6	1.143547381	1.249145239
7	1.091321116	1.238421213
8	1.141508974	1.248493797
9	1.09938961	1.241191156
10	1.149773954	1.268921285
11	1.14069022	1.270229874
12	1.146656754	1.24167543
13	1.102907906	1.259982772
14	1.101953101	1.243815581
15	1.118310246	1.268753067
15	1.118402123	1.266220149
17	1.108553005	1.254641827
18	1.146853532	1.240699792
19	1.10127623	1.276807679
20	1.13689867	1.251805626

APPENDIX B SECS/GEM-AE EQUIP-R PROCESSING TIME

No.	SECS/GEM-AE (AES-GCM-128)	SECS/GEM-AE (AES-GCM-256)
1	0.620277937	0.715758975
2	0.643083313	0.737954427
3	0.650186003	0.701477903
4	0.641501656	0.707161011
5	0.636158371	0.762642091
6	0.623782339	0.696137813
7	0.617995068	0.739759155
8	0.635986709	0.747369606
9	0.640106076	0.774904482
10	0.647909796	0.745524043
11	0.634878678	0.763319636
12	0.628764103	0.78573389
13	0.631992942	0.705708934
14	0.633544788	0.76305516
15	0.617127113	0.75644046
16	0.643456579	0.718080771
17	0.635254461	0.70724062
18	0.637420692	0.755675971
19	0.636070838	0.785254055
20	0.61809492	0.69554342

LIST OF PUBLICATIONS

Jaisan, A., Manickam, S., Laghari, S. A., Rehman, S. U., & Karuppayah, S. (2021). Secured SECS/GEM: A Security Mechanism for M2M Communication in Industry 4.0 Ecosystem. *International Journal of Advanced Computer Science and Applications*. <https://doi.org/10.14569/IJACSA.2021.0120828>